Movable Antenna-Enabled Secure Transmission for Active RIS-Aided ISAC Systems

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***Abstract*—The integrated sensing and communication (ISAC) technology empowered by movable antennas (MAs) has increased the possibility of wireless communication development. In this paper, we investigate a discrete MA-enabled active reconfigurable intelligent surface (RIS)-aided ISAC system. In this system, the position of discrete MAs can be dynamically adjusted to enhance channel conditions, thereby improving the communication security of legitimate users against potential eavesdropping risks from target. In particular, we formulate an optimization problem for the max-min secrecy rate (SR) by jointly optimizing beamforming matrix, phase shift matrix and MA discrete positions. An efficient alternating optimization (AO) based on penalty-successive convex approximation (SCA) and penalty-dual-decomposition (PDD) approach is proposed to obtain a high-quality initial solution for beamforming matrix and phase shift in the non-convex problem. Next, we propose a dueling double deep Q-Network (D3QN)-twin delayed and deep deterministic policy gradient (TD3) method to cope with the hybrid action space issue of the non-convex problem with highly coupled variables. Numerical results demonstrate that the introduction of MAs can effectively improve SR performance compared to fixed-position antenna (FPA) structure in RIS-aided ISAC systems. In addition, the proposed AO-based D3QN-TD3 method validate the suitability and superiority compared to the D3QN-TD3 method without AO algorithm.**

***Index Terms*—Movable antenna, integrated sensing and communication, reconfigurable intelligent surface, deep reinforcement learning.**

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I. Introduction

A. Background

WITH the development of wireless communication in sixth generation (6G) networks, the integrated sensing and communication (ISAC) technology has received increasing attention from both academia and industry, and are considered one of the promising technologies for implementing the next generation of wireless communication systems [1], [2], [3]. Unlike separate communication systems and sensing systems, ISAC systems combine sensing and communication functions in one system through the same spectrum occupation, shared hardware circuits, and unified waveform design [4]. They not only share the same resources, hardware facilities, and signal processing modules, but also have the advantages of low hardware cost, low power consumption, and high spectral efficiency [5], [6]. Most current research is conducted with fixed numbers and positions of base station antennas, which results in wireless networks being unable to flexibly allocate their antenna resources based on the uneven distribution of user space. At the same time, current researches improve performance by using more fixed position antenna (FPA) at the cost of increasing hardware costs and energy consumption. Therefore, FPA structure may not meet the high-performance expectations of future 6G ISAC networks [7].

Compared to traditional FPA structures, movable antenna (MA) technology can flexibly adapt to multipath channel characteristics in dynamic environments by changing the shape, position, polarization, etc. of the transmitting/receiving antenna, resulting in a higher signal-to-noise ratio at the receiving end and improving the reliability of the communication system [8], [9], [10]. Specifically, the concept of MA introduces a new dimension of spatial degrees of freedom (DoF) for wireless communication systems, making beamforming more precise and further improving system spectral and energy efficiency [11]. In addition to improve channel power gain, beamforming and spatial multiplexing of MA systems can provide additional DoF. Especially by jointly designing the position and beamforming of MAs, more flexible beamforming can be achieved [12]. Specifically, optimizing the position of antenna array elements can enhance the correlation between guidance vectors in different

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directions, thereby achieving larger beamforming gains in all directions [13].

For spatial multiplexing, optimizing the position of MA can reshape the channel matrix of MIMO systems, thus further improving channel capacity and spectral efficiency compared to traditional FPA MIMO systems [14]. Ref. [14] confirmed that compared to traditional FPA systems, MA systems can significantly improve the channel capacity of the system by reshaping the channel matrix through optimizing antenna positions. In addition, MAs can suppress multi-user interference by flexibly configuring antenna positions. Reference [15] proposes a new multiple access method, namely fluid antenna multiple access. Specifically, compared to traditional multiple access methods that require complex signal processing and encoding, fluid antenna multiple access only requires moving the antennas of each user to the signal to interference ratio region to effectively suppress multi-user interference without the need for complex signal processing. Furthermore, as space-air-ground integrated communication technology emerges as a key research direction for future communication development, MA technology can adapt to and optimize channel conditions by adjusting antenna positions and angles [16]. This capability enables it to better address the complex and dynamic communication network environments of space-air-ground integration, achieving robust system performance improvements under intricate and variable transmission conditions [17], [18]. Therefore, as one of the key technologies in 6G mobile communications, MA technology can organically integrate with various 6G mobile communication application scenarios. Leveraging its inherent technical advantages, it provides momentum for the development of 6G mobile communication technology, thereby accelerating its rapid advancement and widespread application.

With the ISAC technology being identified as one of the typical application scenarios for 6G mobile networks, MA technology can significantly enhance both communication and sensing performance by leveraging spatial DoF without increasing the number of antennas or radio frequency components, thus demonstrating significant application potential in future 6G networks [19], [20]. In addition, the change in its spatial position effectively increases the antenna aperture, which can improve sensing accuracy. Therefore, it has important application potential in future 6G networks. Research has shown that under given signal-to-noise ratio conditions, the Cramér-Rao bound (CRB) of the mean square error of directional angle estimation is mainly determined by the variance of multiple antenna positions. Therefore, without changing the number of antennas, increasing the range of antenna movement can significantly improve the angle estimation performance [21]. However, due to the different goals of communication and sensing, the position and orientation of the MA needs to be designed reasonably to achieve the best compromise of synesthesia performance.

Moreover, in order to further explore the DoF in the spatial domain to improve communication performance, MA is introduced into Reconfigurable intelligent surfaces (RIS) [22], [23], [24]. By flexibly adjusting the topological structure of RIS reflection units, an intelligent and controllable wireless channel propagation environment can be further achieved [25]. RIS has became a revolutionary method for improving additional DoF due to the ability to intelligently create favorable propagation environments and achieve satisfactory performance [26], [27]. Specifically, Reference [26] considered a scenario where the transmit antenna, receive antenna, and RIS can rotate, derives an upper bound on the system throughput capacity under the Rayleigh channel, and solves for the optimal rotation angles of the transmit/receive antennas and RIS based on this. The results indicated that optimizing the rotation angles of the RIS significantly improves the performance of the communication system. In addition to rotation, Reference [27] analyzed RIS composed of linear movable arrays and points out that by optimizing the positions of the individual elements of the RIS the phase shift effects caused by different transmission angles and arrival angles at the RIS end can be effectively eliminated, significantly improving the system's reachability.

In addition, due to the shared spectrum and broadcast characteristics of wireless transmission, MA-enabled systems face unique security challenges. Undoubtedly, secure communication issues are crucial for the design of MA-enabled systems [28], [29], [30]. To address this issue, Physical Layer Security (PLS) has been used by researchers to handle secure communication problems [31], [32], [33]. PLS is an important communication security technology aimed at enhancing the security of communication systems by utilizing the characteristics of wireless channels and physical layer technology to prevent unauthorized users from eavesdropping, interfering, or tampering with communication data [34], [35], [36], [37]. Compared to traditional cryptographic based security mechanisms, PLS provides an additional security guarantee by directly utilizing the physical characteristics of signal transmission to ensure communication security [35]. Specifically, by optimizing the position of the transmitting antenna and changing the channel state information, the eavesdropper's channel is placed in a deep fading state, thereby enhancing PLS [38].

In summary, to fully leverage the advantages of MA in communication security, we must focus on optimizing the best solution for MA's selectable positions and overcome the limitations of static beamforming design. By developing an efficient and flexible MA position selection scheme, we can not only fully unlock MA's potential to enhance communication capacity but also increase DoF and ensure communication security.

B. Contributions and Organization

Based on the aforementioned discussion, we study the discrete MA-enabled secure transmission for RIS-aided ISAC systems. Unlike the optimization of continuous MA positions, moving discrete antenna positions can reduce the complexity of mechanical systems and facilitate practical deployment. In addition, most optimization schemes were limited to traditional optimization methods and failed to derive solutions that adapt to dynamic environments. To this end, we are committed to designing an efficient joint traditional optimization methods and deep reinforcement learning (DRL) algorithm to address the secure transmission issues in ISAC systems. The main contributions are listed as follows:

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- In this paper, we propose a novel discrete MA-enabled secure transmission for active RIS-aided ISAC systems, where the system firstly considering the innovation of simultaneously optimizing the discrete positions of transmitting and receiving antennas at BS. The max-min secrecy rate (SR) optimization problem is formulated by jointly optimizing the transmit beamforming of BS, the phase shift of RIS and the position of MAs, respectively. Moreover, the communication requirements of users are ensured while the constraints of the CRB for target estimation, power consumption, and the positions of MAs are maintained according to the problem formulation.
- In order to optimize variable processing for complex continuous and discrete actions, we propose an alternating optimization (AO)-based dueling Ddouble deep Q-Network (D3QN)-twin delayed and deep deterministic policy gradient (TD3) method to solve the established long-term planning problem. Specifically, to address the tightly coupled challenges of multi-antenna array positioning, transmit beamforming, and phase shifting, the AO-based D3QN-TD3 secure transmission scheme ensures the security of the ISAC system by learning effective strategies.
- We construct the AO-based algorithm to optimize both transmit beamforming and the reflection coefficient of the RIS. Specifically, the transmit beamforming is optimized by adopting the penalty-successive convex approximation (SCA) method to relax rank-one constraint. Moreover, we develop a penalty-dual-decomposition (PDD) algorithm to deal with the complex quartic expressions in RIS phase shift optimization.
- The comprehensive simulation results demonstrate benefits of the MA-enable ISAC systems, which indicate that the MA-enabled ISAC system outperforms the FPA-based ISAC system due to the spatial diversity gain and additional DoF introduced by antenna position optimization. In addition, expanding the MA moving region area increases the equivalent array aperture, thereby effectively extending the coverage range of the ISAC system without adding more antennas. Furthermore, compared to the traditional D3QN-TD3 approach, the AO-based D3QN-TD3 method achieves significant improvements in safety performance across all aspects.

C. Organization and Notation

The remainder of the paper is organized as follows. The related work is comprehensively described in Section II. In Section III, we present the MA-enabled Active RIS-aided secure ISAC system and the related problem formulation. Section IV provides the efficient methods to solve the max-min SR optimization problem. The simulation results demonstrate the effectiveness of the proposed method are presented in Section V. Lastly, Section VI provides the concluding remarks of this paper.

Notation: In this paper, scalars, vectors, and matrices are signified by standard fonts, bold lowercase letters, and bold uppercase letters, respectively. The conjugate, transpose, Hermitian transpose, and vectorization operations are denoted by $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$, and $\mathrm{vec}(\cdot)$, respectively. $|\cdot|$, $\|\cdot\|$, and $\|\cdot\|_F$ stand for the norm, l_2-norm, and Frobenius-norm, respectively. A positive semi-definite matrix is represented by $\mathbf{A} \succeq \mathbf{0}$. $\Re\{\cdot\}$ and $\Im\{\cdot\}$ indicate the real and imaginary components of a complex value, respectively. $\otimes$ stands for the Kronecker product, $\mathbf{I}_N$ corresponds to a N-dimensional identity matrix, and $\mathrm{diag}(\mathbf{a})$ denotes a diagonal matrix. In addition, $\mathcal{CN}(0, \sigma^2)$ refers to the complex normal distribution with zero-mean and variance σ^2, and $\mathbb{E}\{\cdot\}$ signifies the expectation operator. $[\cdot]^+$ is a value greater than or equal to zero.

II. Related Work

In this section, we discuss the current state of MA technology research with a focus on MA techniques. First, we analyze the significant potential for MA technology to integrate with ISAC and RIS technologies, highlighting its advantages over FPA. Additionally, we examine the security implications of MA technology applications and the need for specific optimization methods to address these challenges.

The current ISAC research is usually conducted based on FPA, and considerable exploration and discovery have been made from RIS technology and safety considerations. Zhu et al. [39] have applied a deep deterministic policy gradient (DDPG) and soft actor-critic (SAC) algorithm to maximize the long-term average security rate in the simultaneous transmitting and reflecting RIS (STAR-RIS) aided ISAC secure system. Kumar et al. [40] addressed a beampattern maximization problem in a secure active RIS-enabled ISAC system via a high-performance solution. Liu et al. [41] adopted a DRL algorithm to maximize the achievable SR of all the legitimate users in a RIS-assisted ISAC system. Chu et al. [42] leveraged the RIS to assist legitimate wireless communication while enhancing the sensing performance in a secure RIS-aided ISAC system. Guo et al. [43] have achieved a high-quality sensing beampattern while meeting covertness and security communication requirements in a STAR-RIS assisted ISAC system. Yao et al. [44] proposed a novel secure scheme for transmitting confidential information and performing radar sensing in the RIS-enabled vehicle-to-everything networks. Li et al. [45] ignored the issue of internal eavesdropping and assume that legitimate users are not malicious. Yang et al. [46] focused on security beamforming technology in RIS-assisted ISAC systems, with an emphasis on sensing accuracy and eavesdropper localization. Yang et al. [47] proposed a novel continuous interference cancellation scheme that ensures the integrity of legitimate user links while utilizing radar signals to disrupt potential eavesdropping behavior. Zhang et al. [48] investigated an active RIS-assisted ISAC system to maximize the SR by adopting the alternating optimization algorithm. However, the aforementioned studies are all based on traditional FPA structure, which relies on increasing the number of antennas and radio frequencies to enhance service quality, leading to significant increases in hardware costs and power consumption. By leveraging spatial DoF, MA technology can substantially improve communication and sensing performance without adding more antennas or RFs, thus holding significant application potential in future 6G networks.

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The introduction of MA technology provides a new paradigm for the development of future communication. For secure communication, the existing MA technologies are mainly studied in combination with PLS, like direction modulation and covert communication. By optimizing the position of the antenna, potential better channels can be selected for data transmission, thereby enhancing the secure transmission rate. The paper proposed a joint optimization scheme for transmit precoding and MA positioning to maximize SR in MIMO systems, demonstrating significant security improvements over FPAs. In [49], a joint optimization scheme for transmit precoding, artificial noise, and MA positioning was proposed to maximize SR in MIMO systems, demonstrating significant security improvements over FPAs. Reference [50] demonstrated that MAs can significantly expand the achievable SR region for physical-layer service integration compared to FPAs through joint beamforming and antenna positioning optimization. For the position optimization of MAs, in [51], [52], and [53], the gradient descent algorithm, particle swarm optimization algorithm, and block coordinate descent algorithm were respectively proposed to enhance communication security performance. In [54], considering hardware damage and imperfect channel estimation, a uniform packet and non-uniform packet scheme was proposed to optimize the position of the MA, achieving higher security performance than the traditional FPAs under discrete position constraints.

However, the aforementioned works have studied the continuously adjustable positions of MAs in the ISAC system. For example, Zhang et al. [55] addressed the optimization of antenna positions in a continuous manner, and demonstrated that under given antenna counts and power budgets, even minor adjustments to the position of a mobile antenna array can significantly enhance the performance of integrated communication and sensing systems. Furthermore, Lyu et al. [56] further investigated the joint design of continuous antenna position and beamforming in a MA-aided ISAC system, aiming to maximize the weighted sum of communication rate and sensing mutual information under a given total transmit power budget. Considering practical scenarios for ISAC, enhancing secure communication and target sensing through the optimization of MAs under discrete position constraints is a key issue [57]. Some studies have attempted to address the issue of MA discretization in position selection [58]. Shao et al. [58] proposed six-dimensional (6D) position and rotation of 6D MA surface to maximize network capacity. Zou et al. [59] achieved a 33% reduction in transmission power compared to traditional FPAs by jointly optimizing the discrete positions and beamforming of a one-dimensional MA, while meeting communication and sensing performance requirements. Wang et al. [60] has further investigated a two-dimensional MA-assisted pan-sensing integrated system. By employing deep reinforcement learning to jointly optimize discrete antenna positions and precoding schemes, it significantly enhances multi-user communication rates while satisfying sensing requirement constraints. Mei et al. [61] transformed the continuous antenna position optimization problem into a discrete sampling point selection problem based on point by point channel information, and developed a graph based optimization technique to obtain the optimal solution.

For the ISAC system assisted by MAs, the trade-off between communication and sensing has been studied. By optimizing the positions of the MAs, existing works have shown that the performance of communication and sensing can be enhanced. In [62], by minimizing the CRB for sensing under communication constraints, the MA-based ISAC system achieves substantial performance gains over conventional FPA setups. Pang et al. [63] revealed the potential uses of movable components through performance analysis, and then designed a joint beamforming and component position optimization scheme, proving that this scheme can improve communication rate and sensing accuracy. Ding et al. [64] proposed a dual layer random position algorithm to solve the problem of maximizing the weighted sum of sensing rate and communication rate in MA assisted systems. Qin et al. [65] formulated the CRB minimization problem in the MA-aided multiuser ISAC system. Xiu et al. [66] aimed to maximize capacity by jointly optimizing MA positions and beamforming in the MA-assisted low-altitude vehicle ISAC model. Fang et al. [67] has provided a new definition for the mode of antenna selection in the simultaneous sensing and backscatter communication in cluttered environments. In addition, interesting technologies for the assistance of MAs considering safety performance in ISAC systems have been studied. In [68], a proximal policy optimization algorithm was proposed to maximize the SR while ensuring sensing performance in the MA-aided ISAC systems under imperfect eavesdropper's Channel State Information. Employing a two-layer penalty-based algorithm, the proposed MA-assisted RIS-ISAC system in [69] is shown to significantly outperform conventional FPA-based systems in enhancing secure performance by jointly optimizing beamforming, RIS coefficients, and MA positions.

In addition, the RIS assisted MA system can significantly enhance the transmission capacity of existing communication systems, providing a new paradigm for wireless communication in complex dynamic scenarios. Lyu et al. [70] firstly designed a robust transmission scheme by combining RIS and MA to maximize the primary rate while ensuring the secondary transmission quality. Compared to traditional RIS assisted FPAs, [71] revealed that MA enabled RIS can optimize antenna position to improve channel capacity. Wei et al. [72] has provided the conclusion that the performance gain of MA relative to traditional FPAs may decrease in the presence of RIS passive beamforming. Hu et al. [27] designed the optimal non-uniform discrete phase shift for RIS, which can offset the phase distribution between different levels of channels. Of course, the combination of MA and RIS also provides a fresh approach to in-depth research on ISAC systems. A novel RIS structure combining MA under spherical wave propagation conditions has been proposed in [73], which optimizes the weighted sum rate of communication users while ensuring perceptual performance. Wu et al. [74] studied a two-dimensional (2D) MA array and RIS assisted ISAC system to address the issue of communication and sensing performance in dead zones.

Although all current MAs utilize micro-electromechanical systems or stepper motors to achieve high precision, miniaturization, and low power consumption, in practical applications, the movement and repositioning of MAs require additional

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power to operate mechanical components, which undoubtedly leads to increased energy consumption. Furthermore, all existing studies assume that the latency of MA movement is tolerable and negligible. However, for broadband systems or even large transmitter area systems, the latency of MA movement can also impose limitations on system performance improvement. Therefore, research on the latency and energy efficiency issues of MA devices is an unavoidable challenge for the practical application of MA technology in future mobile communication scenarios. Wei et al. [75] firstly proposed a detailed power consumption model for stepper motor-driven multi-MA systems based on fundamental motor theory, which accounts for the nonlinear relationship between mechanical energy consumption and movement speed, and apply it to the joint optimization problem for energy efficiency maximization. By analyzing the actual time delay and energy consumption caused by antenna movement, Ding et al. [76] investigated the energy efficiency maximization problem in MA-assisted multi-user uplink communication systems, derives the upper bound of energy efficiency, and provides important references for practical system deployment. Li et al. [77] explicitly focused on MA movement delay and finds that the movement delay decreases as the maximum speed of MA increases. However, under the same speed constraints, by jointly optimizing matching and trajectory, the time wasted due to avoidance and unreasonable scheduling can be significantly reduced. Wei et al. [78] specifically established a mechanical power consumption model for the single MA system, and demonstrated that MA can still outperform traditional FPA systems even when accounting for additional power consumption, by jointly optimizing the MA's position, moving speed, and transmit power to maximize energy efficiency. Chen et al. [79] demonstrated that flexibly adjusting antenna positions to exploit spatial DoF can effectively enhance desired signals and suppress interference. Furthermore, when the antenna movement area expands beyond a certain size, the improvement in system energy efficiency tends to saturate.

It is also concerning that robust optimization of MA positions based on traditional optimization methods struggles to handle fast-paced dynamic scenarios due to multiple uncertainties. Furthermore, it is difficult to efficiently process high-dimensional actions and complex state spaces. Artificial intelligence-assisted technologies such as DRL are also being vigorously applied to MA-assisted communication systems. An AO-based deep Q-network (DQN) algorithm method was proposed in [80] to design the moving trajectories of MAs, achieving a covert rate improvement. Weng et al. [81] proposed a multi agent deep deterministic policy gradient (MADDPG) algorithm to optimize real-time transmission beamforming and antenna movement strategies, maximizing the sum rate under imperfect channel state information. Li et al. [82] proposed a multi-agent twin delayed deep deterministic policy gradient method to maximize the sum of achievable rates for all users in the MA-enabled cell-free communication network system. Xie et al. [83] proposed a heterogeneous multi-agent deep deterministic policy gradient algorithm to manage beamforming and antenna positioning tasks in MA-aided systems. Shao et al. [84] proposed a low-complexity Actor-Critic (A2C) based DRL algorithm to address the issue of coupled position and rotation constraints attributed to 6DMA.

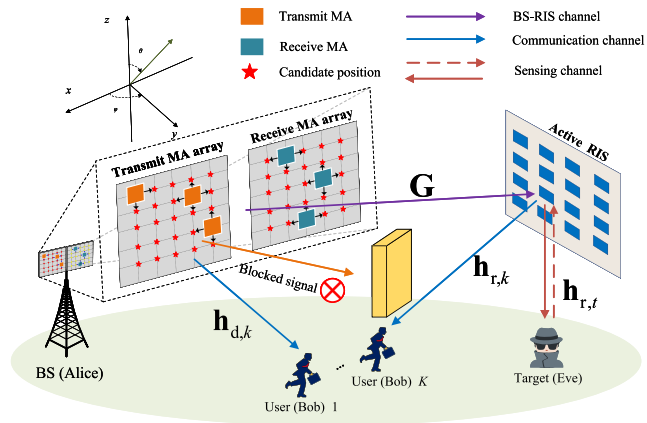

Fig. 1. Illustration of the MA-enabled active RIS-aided ISAC model.

Table I summarizes the classic literature comparisons to our work.

III. System Model

In this paper, we consider a MA-enabled active RIS-aided ISAC system as illustrated in Fig. 1, where a dual-function BS employs two MA uniform planner array (UPA) for transmitting and receiving signals. Specifically, BS transmits information to K single-antenna users, while a single-antenna eavesdropping target to be sensed is hovering nearby to intercept their data. The M discrete transmit and N discrete receive MAs can be adjusted simultaneously within a given 2D transmit and receive area, respectively. Assuming the direct link from BS to the eavesdropping target is unavailable due to blockages, the active RIS is responsible for establishing a robust virtual line-of-sight (LoS) link to ensure communication and sensing performance. To maximize the potential of the MA-enabled system under consideration, the perfect channel state information regarding the transmitter region is assumed achievable at BS.

A. Signal Model

The transmitted signal incorporating both communication and radar functions in the t-th is represented as follows

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{W}_{\mathrm{c}}\mathbf{s}_{\mathrm{c}}(t) + \mathbf{W}_{\mathrm{r}}\mathbf{s}_{\mathrm{r}}(t)\\ &= [\mathbf{W}_{\mathrm{c}}, \quad \mathbf{W}_{\mathrm{r}}][\mathbf{s}_{\mathrm{c}}^T(t), \quad \mathbf{s}_{\mathrm{r}}^T(t)]^T = \mathbf{W}\mathbf{s}(t), \qquad (1)\end{aligned}$$

where $\mathbf{s}_{\mathrm{c}} = [s_{\mathrm{c},1}, \ldots, s_{\mathrm{c},K}]^T \in \mathbb{C}^{K\times 1}$ denotes the communication symbols intended for K users, and $\mathbf{s}_{\mathrm{r}} = [s_{\mathrm{r},1}, \ldots, s_{\mathrm{r},M}]^T \in \mathbb{C}^{M\times 1}$ denotes M individual radar waveforms. $\mathbf{W}_{\mathrm{c}} = [\mathbf{w}_{\mathrm{c},1}, \mathbf{w}_{\mathrm{c},2}, \ldots, \mathbf{w}_{\mathrm{c},K}] \in \mathbb{C}^{M\times K}$ and $\mathbf{W}_{\mathrm{r}} = [\mathbf{w}_{\mathrm{r},1}, \mathbf{w}_{\mathrm{r},2}, \ldots, \mathbf{w}_{\mathrm{r},M}] \in \mathbb{C}^{M\times M}$ represent the corresponding communication and radar beamforming matrix, respectively. In addition, we define the combined beamforming matrix $\mathbf{W} \triangleq [\mathbf{W}_{\mathrm{c}}, \mathbf{W}_{\mathrm{r}}] \in \mathbb{C}^{M\times(K+M)}$ and symbol vector $\mathbf{s}(t) \triangleq [\mathbf{s}_{\mathrm{c}}^T(t), \mathbf{s}_{\mathrm{r}}^T(t)]^T$ to simplify the analysis.

When the frame length T is assumed to be sufficiently large, the sample covariance matrix of the transmitted signal $\mathbf{x}$ can be

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TABLE I
COMPARISONS WITH RELATED WORKS

Related works	MA	RIS	Communication indicator	Sensing indicator	Method
[39]	×	STAR-RIS	SR	SINR	DDPG and SAC
[43]	×	STAR-RIS	Covert communication	Beampattern	semidefinite relaxation (SDR)
[48]	×	Active RIS	SR	SINR	SDR and majorization-minimization (MM)
[55]	Continuous	×	Communication rate	Detection	Proximal distance and extrapolated projected gradient
[57]	Discrete	×	SINR	×	Generalized Bender's decomposition
[60]	Discrete	×	Communication rate	SINR	Actor and critic algorithm
[62]	Continuous	×	SNR	CRB	MM based Rosen's gradient projection
[68]	Continuous	×	SR	SINR	Proximal policy optimization
[69]	Continuous	Passive RIS	Communication rate	SNR	Two-layer penalty-based algorithm
[74]	Continuous	Passive RIS	Communication rate	Beampattern	SCA and SDR
[81]	Continuous	×	Communication rate	×	MADDPG
Ours	Discrete	Active RIS	SR	CRB	AO-based D3QN-TD3

regarded as identical to its statistical covariance matrix

$$\mathbf{R} \triangleq \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}(t)\mathbf{x}^H(t) \approx \mathbb{E}(\mathbf{x}(t)\mathbf{x}^H(t)) = \sum_{k=1}^{K}\mathbf{R}_k + \mathbf{R}_{\mathrm{r}}, \tag{2}$$

where $\mathbf{R}_k = \mathbf{w}_{\mathrm{c},k}\mathbf{w}_{\mathrm{c},k}^H$ and $\mathbf{R}_{\mathrm{r}} = \mathbf{W}_{\mathrm{r}}\mathbf{W}_{\mathrm{r}}^H$.

B. Channel Model

The positions of transmit and receive MAs with two minimum spacing of $D_{1,\min}$ and $D_{2,\min}$ are adjusted within the 2D movable range $\mathcal{C}_t$ and $\mathcal{C}_r$, respectively. We define the number of I and J candidate discrete positions of the transmit and receive MAs in $\mathcal{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_I\}$ and $\mathcal{Q} = \{\mathbf{q}_1, \ldots, \mathbf{q}_J\}$, where $\mathbf{p}_i = [p_{i,x}, p_{i,y}]^T$ and $\mathbf{q}_j = [q_{j,x}, p_{j,y}]^T$ denote the i-th and j-th candidate position, $i \in \mathcal{I}$ and $j \in \mathcal{J}$. In addition, the number of transmit paths for the direct channel from the BS to the k-th user is defined as L_t^k. The antenna positioning vector of M transmit MAs and N receive MAs are described by $\tilde{\mathbf{t}} = [\mathbf{t}_1, \ldots, \mathbf{t}_M]^T \in \mathbb{R}^{M\times 2}$ and $\tilde{\mathbf{r}} = [\mathbf{r}_1, \ldots, \mathbf{r}_N]^T \in \mathbb{R}^{N\times 2}$, where $\mathbf{t}_m = [x_m, y_m]^T \in$ and $\mathbf{r}_n = [x_n, y_n]^T$ for $m \in \mathcal{M}$ and $n \in \mathcal{N}$, respectively. The direct channel vector between the m-th MA and K users is denoted as

$$\mathbf{h}_m(\mathbf{t}_m) = [h_{m,1}(\mathbf{t}_m), \ldots, h_{m,K}(\mathbf{t}_m)]^T \in \mathbb{C}^K, \tag{3}$$

where $h_{m,k}(\mathbf{t}_m)$ is the channel coefficient between the m-th MA and the k-th user. Next, we define a matrix $\tilde{\mathbf{H}}_m = [\mathbf{h}_m(\mathbf{p}_1), \ldots, \mathbf{h}_m(\mathbf{p}_I)] \in \mathbb{C}^{K\times I}$ to collect the direct channel vectors from the m-th MA to all K users for all M candidate discrete MA locations. The channel coefficient $h_{m,k}(\mathbf{p}_i)$ between the m-th MA element at $\mathbf{p}_i$ and the k-th user is modeled as follows,

$$h_{m,k}(\mathbf{p}_i) = \mathbf{1}_{L_t^k}^H \mathbf{\Sigma}_k \mathbf{h}_k(\mathbf{p}_i), \tag{4}$$

where $\mathbf{\Sigma}_k \in \mathbb{C}^{L_r^k \times L_t^k}$ is the path response matrix of L_t^k transmit paths and L_r^k receive paths from the BS to the k-th user. $\mathbf{h}_k(\mathbf{p}_i)$ denotes the transmit field response vector between the k-th user and the MA at position $\mathbf{p}_i$, which is given by $\mathbf{h}_k(\mathbf{p}_i) = [e^{j\frac{2\pi}{\lambda}\rho_{k,1}(\mathbf{p}_i)}, \ldots, e^{j\rho_{k,L_p}(\mathbf{p}_i)}]^T$, where $\rho_{k,l_p}(\mathbf{p}_i) = x\sin\theta_{k,l_t}\cos\phi_{k,l_t} + y\cos\theta_{k,l_t}$ represents the phase difference of the l_p-th channel path between antenna position $\mathbf{p}_i$ and the first position $\mathbf{p}_1$. Furthermore, due to the positions of RIS elements and antennas of the users are fixed, we define $\mathbf{h}_{\mathrm{r},k} \in \mathbb{C}^L$ is the channel vector between RIS and k-th user. According to $\tilde{\mathbf{H}}_m$, $\mathbf{h}_m(\mathbf{t}_m)$ can be formulated as

$$\mathbf{h}_m(\mathbf{t}_m) = \tilde{\mathbf{H}}_m \mathbf{a}_m, \tag{5}$$

where $\mathbf{a}_m \triangleq [a_{m,1}, \ldots, a_{m,I}]^T$. Here, $a_{m,i} \in \{0, 1\}$ with $\sum_{i=1}^{I} a_{m,i} = 1$ is a binary number used to indicate whether the m-th candidate position is selected as a transmit MA position. For the considered MA-enabled system, the direct channel matrix between the BS and users is given by

$$\mathbf{H} = \tilde{\mathbf{H}}\mathbf{A} = [\mathbf{h}_1(\mathbf{t}_1), \ldots, \mathbf{h}_M(\mathbf{t}_M)] \in \mathbb{C}^{K\times M}, \tag{6}$$

where

$$\begin{aligned}\tilde{\mathbf{H}} &= [\tilde{\mathbf{H}}_1, \ldots, \tilde{\mathbf{H}}_M] \in \mathbb{C}^{K\times MI} \\ &= [\mathbf{h}_{\mathrm{d},1}, \ldots, \mathbf{h}_{\mathrm{d},K}]^T,\end{aligned} \tag{7}$$

where $\mathbf{h}_{\mathrm{d},k} \in \mathbb{C}^{MI}$ is the k-th row of channel matrix of $\tilde{\mathbf{H}}$ and the transmit MA position selection matrix $\mathbf{A}$ is defined as follows

$$\mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{0}_I & \mathbf{0}_I & \cdots & \mathbf{0}_I \\ \mathbf{0}_I & \mathbf{a}_2 & \mathbf{0}_I & \cdots & \mathbf{0}_I \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \mathbf{0}_I & \mathbf{0}_I & \mathbf{0}_I & \cdots & \mathbf{a}_M \end{bmatrix} \in \mathbb{C}^{MI\times M}. \tag{8}$$

Similarly, the channel matrix of the BS-RIS transmitting and RIS-BS receiving path are expressed as follows, respectively,

$$\mathbf{G}_t = \tilde{\mathbf{G}}_t\mathbf{A} = [\mathbf{g}_{t,1}(\mathbf{t}_1), \ldots, \mathbf{g}_{t,M}(\mathbf{t}_M)] \in \mathbb{C}^{L\times M}, \tag{9}$$

$$\mathbf{G}_r = \tilde{\mathbf{G}}_r\mathbf{B} = [\mathbf{g}_{r,1}(\mathbf{r}_1), \ldots, \mathbf{g}_{r,N}(\mathbf{r}_N)] \in \mathbb{C}^{L\times N}, \tag{10}$$

where $\tilde{\mathbf{G}}_t = [\tilde{\mathbf{G}}_{t,1}, \ldots, \tilde{\mathbf{G}}_{t,M}] \in \mathbb{C}^{L\times MI}$ with $\tilde{\mathbf{G}}_{t,m} = [\mathbf{g}_{t,m}(\mathbf{p}_1), \ldots, \mathbf{g}_{t,m}(\mathbf{p}_I)] \in \mathbb{C}^{L\times I}$ to denote the channel vectors from m-th MA to the active RIS for all candidate discrete transmit MA locations. The channel coefficient $\mathbf{g}_{t,m}(\mathbf{p}_i)$ between the m-th MA element at $\mathbf{p}_i$ and the RIS is

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modeled as follows

$$\mathbf{g}_{t,m}^H(\mathbf{p}_i) = \mathbf{1}_{L_t}^H \mathbf{\Sigma}_1 \mathbf{G}(\mathbf{p}_i), \tag{11}$$

where $\mathbf{G}(\mathbf{p}_i) \in \mathbb{C}^{L_t \times L}$ denotes the field response vector between the MA at position $\mathbf{p}_i$ and RIS, $\mathbf{\Sigma}_1 \in \mathbb{C}^{L_r \times L_t}$ is the path response matrix of all transmit paths from the BS to RIS, where L_t and L_r the numbers of transmit and receive paths of the BS-RIS channel, respectively.

In addition, $\tilde{\mathbf{G}}_r = [\tilde{\mathbf{G}}_{r,1}, \ldots, \tilde{\mathbf{G}}_{r,N}] \in \mathbb{C}^{L \times NJ}$ with $\tilde{\mathbf{G}}_{r,n} = [\mathbf{g}_{r,n}(\mathbf{q}_1), \ldots, \mathbf{g}_{r,n}(\mathbf{q}_J)] \in \mathbb{C}^{L \times J}$ is denoted as the channel vectors from n-th MA to the active RIS for all candidate discrete receive MA locations. The channel coefficient $\mathbf{g}_{r,n}(\mathbf{q}_j)$ between the n-th MA element at $\mathbf{q}_j$ and the RIS is modeled as $\mathbf{g}_{r,n}^H(\mathbf{q}_j) = \mathbf{1}_{L_r}^H \mathbf{\Sigma}_2 \mathbf{G}(\mathbf{q}_j)$, where $\mathbf{\Sigma}_2 \in \mathbb{C}^{L_t \times L_r}$ and $\mathbf{G}(\mathbf{q}_j) \in \mathbb{C}^{L_r \times L}$ denote the path response matrix of all transmit paths from the BS to RIS and the field response vector from all MAs to RIS, respectively. The receive MA position selection matrix $\mathbf{B}$ is defined as follows

$$\mathbf{B} = \begin{bmatrix} \mathbf{b}_1 & \mathbf{0}_J & \mathbf{0}_J & \cdots & \mathbf{0}_J \\ \mathbf{0}_J & \mathbf{b}_2 & \mathbf{0}_J & \cdots & \mathbf{0}_J \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \mathbf{0}_J & \mathbf{0}_J & \mathbf{0}_J & \cdots & \mathbf{b}_N \end{bmatrix} \in \mathbb{C}^{NJ \times N}, \tag{12}$$

where $\mathbf{b}_n \triangleq [b_{n,1}, \ldots, b_{n,J}]^T$ with $\sum_{j=1}^{J} b_{n,j} = 1$ is a binary number used to indicate whether the n-th candidate position is selected as a receive MA position.

C. Communication and Radar Sensing Metrics

The performance of communication systems and radar sensing is significantly affected by transmission rate and target estimation accuracy, respectively. The received signal at the k-th communication user is presented as

$$y_{\mathrm{c},k}(t) = (\mathbf{h}_{\mathrm{d},k}^H + \mathbf{h}_{\mathrm{r},k}^H \mathbf{\Theta} \tilde{\mathbf{G}}_t)\mathbf{A}\mathbf{x}(t) + \mathbf{h}_{\mathrm{r},k}^H \mathbf{\Theta} \mathbf{z}_0(t) + n_k(t), \tag{13}$$

where $\mathbf{\Theta} \in \mathbb{C}^{L \times L}$ denotes the active RIS reflection beamforming matrix with $\mathbf{\Theta} \triangleq \mathrm{diag}\{\boldsymbol{\theta}\}$. $\boldsymbol{\theta} \triangleq [\beta_1 e^{j\varphi_1}, \ldots, \beta_L e^{j\varphi_L}] \in \mathbb{C}^{L \times 1}$ is the reflection coefficient vector with adjustable amplitude $\beta_l \in [0, \beta_{\max}]$ and phase $\varphi_l \in [0, 2\pi)$ of RIS, $\forall l \in \mathcal{L}$, respectively, where $\beta_{\max}$ denotes the maximum amplifying threshold. In addition, $\mathbf{z}_0 \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{z}}^2 \mathbf{I}_L)$ and $n_k \sim \mathcal{CN}(0, \sigma_k^2)$ stands for the additive white Gaussian noise (AWGN) at the active RIS and the k-th user, respectively. Based on the received signal in (13), the SINR for the k-th communication user is

$$\begin{aligned} \gamma_{\mathrm{c},k} &= \frac{|\mathbf{h}_k^H \mathbf{A} \mathbf{w}_k|^2}{\sum_{k' \neq k}^{K+M} |\mathbf{h}_k^H \mathbf{A} \mathbf{w}_{k'}|^2 + \|\mathbf{h}_{\mathrm{r},k}^H \mathbf{\Theta}\|^2 \sigma_{\mathrm{z}}^2 + \sigma_k^2} \\ &= \frac{\mathbf{h}_k^H \mathbf{A} \mathbf{R}_k \mathbf{A}^H \mathbf{h}_k}{\mathbf{h}_k^H \mathbf{A}(\mathbf{R} - \mathbf{R}_k)\mathbf{A}^H \mathbf{h}_k + \|\mathbf{h}_{\mathrm{r},k}^H \mathbf{\Theta}\|^2 \sigma_{\mathrm{z}}^2 + \sigma_k^2}, \end{aligned} \tag{14}$$

where $\mathbf{h}_k^H \triangleq \mathbf{h}_{\mathrm{d},k}^H + \mathbf{h}_{\mathrm{r},k}^H \mathbf{\Theta} \tilde{\mathbf{G}}_t$.

We consider that the two-dimensional DoA estimation of the eavesdropper is imperfect, i.e., $\theta_{\mathrm{h}} \triangleq [\hat{\theta}_{\mathrm{h}} - \Delta\theta_{\mathrm{h}}, \hat{\theta}_{\mathrm{h}} + \Delta\theta_{\mathrm{h}}]$ and $\theta_{\mathrm{v}} \triangleq [\hat{\theta}_{\mathrm{v}} - \Delta\theta_{\mathrm{v}}, \hat{\theta}_{\mathrm{v}} + \Delta\theta_{\mathrm{v}}]$, where θ_{h} and θ_{v} denote the horizontal and vertical DoA estimation and $\Delta\theta_{\mathrm{h}}$ and $\Delta\theta_{\mathrm{v}}$ are the error of estimation, respectively. Note that without any prior information, we cannot effectively distinguish legitimate targets and potential eavesdroppers. We introduce $\mathbf{a}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}})$ to denote the array steering vector between the active RIS and the target. Based on the assumption of plane wave propagation model, the array steering vector is modeled as

$$\mathbf{a}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) = \exp\left(-j[\mathbf{r}_x, \mathbf{r}_y, \mathbf{r}_z]\boldsymbol{\nu}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}})\right), \tag{15}$$

where $[\mathbf{r}_x, \mathbf{r}_y, \mathbf{r}_z] \in \mathbb{R}^{L \times 3}$ denotes the Cartesian coordinates of the RIS elements and $\boldsymbol{\nu}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) = \frac{2\pi}{\lambda}[\cos\theta_{\mathrm{h}}\cos\theta_{\mathrm{v}}, \sin\theta_{\mathrm{h}}\cos\theta_{\mathrm{v}}, \sin\theta_{\mathrm{v}}]^T$ is the wave-number vector with wavelength λ.

The channel between the RIS and the target is modeled as the LoS channel and can be expressed as

$$\mathbf{h}_{\mathrm{r,t}} \triangleq \sqrt{\alpha_t}\mathbf{a}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}), \tag{16}$$

where $\alpha_t \sim \mathcal{CN}(0, \sigma_{\mathrm{t}}^2)$ denotes the complex-valued channel coefficient dependent on the target's radar cross section (RCS) and the round-trip path loss of the RIS-target-RIS link. During the target sensing, communication signals can also be used as sensing signals. The BS works in full duplex mode and receives the echo signal of the target through the BS-RIS-target-RIS-BS in four hops. It should be noted that the noise signal $\mathbf{z}_0(t)$ is ignored due to its extremely weak power when it reaches the base station through multi-stage attenuation. Therefore, the received echo signal by the BS at the t-th time slot is denoted as

$$\begin{aligned} \mathbf{y}_{\mathrm{r}}(t) = &\alpha_t \mathbf{B}^H \tilde{\mathbf{G}}_r^H \mathbf{\Theta}^H \mathbf{\Lambda}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) \mathbf{\Theta} \tilde{\mathbf{G}}_t \mathbf{A}\mathbf{x}(t) \\ &+ \mathbf{B}^H \tilde{\mathbf{G}}_r^H \mathbf{\Theta}^H \mathbf{z}_1(t) + \mathbf{n}_{\mathrm{r}}(t), \end{aligned} \tag{17}$$

where $\mathbf{\Lambda}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) \triangleq \mathbf{a}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}})\mathbf{a}^H(\theta_{\mathrm{h}}, \theta_{\mathrm{v}})$, $\mathbf{z}_1 \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{z}}^2 \mathbf{I}_L)$ and $\mathbf{n}_{\mathrm{r}} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{r}}^2 \mathbf{I}_N)$ are AWGN at the active RIS and the BS, respectively. For the Eve, the received signal is modeled as

$$y_{\mathrm{e}}(t) = \mathbf{h}_{\mathrm{r,t}}^H \mathbf{\Theta}(\tilde{\mathbf{G}}_t \mathbf{A}\mathbf{x}(t) + \mathbf{z}_0(t)) + n_{\mathrm{e}}(t), \tag{18}$$

where $n_{\mathrm{e}} \sim \mathcal{CN}(0, \sigma_{\mathrm{e}}^2)$ denotes the AWGN at Eve. We assume that the sensing signal is unknown to the target. Correspondingly, the received SINR for the Eve is expressed as

$$\begin{aligned} \gamma_{\mathrm{e},k} &= \frac{|\mathbf{h}_{\mathrm{r}}^H \mathbf{A} \mathbf{w}_k|^2}{\sum_{k' \neq k}^{K+M} |\mathbf{h}_{\mathrm{r}}^H \mathbf{A} \mathbf{w}_{k'}|^2 + \|\mathbf{h}_{\mathrm{r,t}}^H \mathbf{\Theta}\|^2 \sigma_{\mathrm{z}}^2 + \sigma_{\mathrm{e}}^2} \\ &= \frac{\mathbf{h}_{\mathrm{r}}^H \mathbf{A} \mathbf{R}_k \mathbf{A}^H \mathbf{h}_{\mathrm{r}}}{\mathbf{h}_{\mathrm{r}}^H \mathbf{A}(\mathbf{R} - \mathbf{R}_k)\mathbf{A}^H \mathbf{h}_{\mathrm{r}} + \|\mathbf{h}_{\mathrm{r,t}}^H \mathbf{\Theta}\|^2 \sigma_{\mathrm{z}}^2 + \sigma_{\mathrm{e}}^2}, \end{aligned} \tag{19}$$

where $\mathbf{h}_{\mathrm{r}}^H \triangleq \mathbf{h}_{\mathrm{r,t}}^H \mathbf{\Theta} \tilde{\mathbf{G}}_t \mathbf{A}$. Based on (14) and (19), the secure rate of k-th Bob can be expressed as follows

$$\begin{aligned} R_{\mathrm{sec},k} &= [R_{\mathrm{c},k} - R_{\mathrm{e},k}]^+ \\ &= [\log_2(1 + \gamma_{\mathrm{c},k}) - \log_2(1 + \gamma_{\mathrm{e},k})]^+. \end{aligned} \tag{20}$$

By stacking T samples, the combined received signals at BS is expressed as

$$\mathbf{Y}_{\mathrm{r}} = \alpha_t \mathbf{F}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}})\mathbf{W}\mathbf{S} + \mathbf{B}^H \tilde{\mathbf{G}}_r^H \mathbf{\Theta}^H \mathbf{Z}_1 + \mathbf{N}_{\mathrm{r}}, \tag{21}$$

where $\mathbf{F}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) \triangleq \mathbf{B}^H \tilde{\mathbf{G}}_r^H \mathbf{\Theta}^H \mathbf{\Lambda}(\theta_{\mathrm{h}}, \theta_{\mathrm{v}}) \mathbf{\Theta} \tilde{\mathbf{G}}_t \mathbf{A}$, and the symbol/noise matrices as $\mathbf{S} \triangleq [\mathbf{s}(1), \ldots, \mathbf{s}(T)]$, $\mathbf{Z}_0 \triangleq [\mathbf{z}_0(1), \ldots, \mathbf{z}_0(T)]$, $\mathbf{Z}_1 \triangleq [\mathbf{z}_1(1), \ldots, \mathbf{z}_1(T)]$ and $\mathbf{N}_{\mathrm{r}} \triangleq [\mathbf{n}_{\mathrm{r}}(1), \ldots, \mathbf{n}_{\mathrm{r}}(T)]$, respectively. The entire first term in (21), including

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both the communication and sensing components, is regarded as a useful signal for estimating target parameters, while the second and third terms are considered as noise signals. The vectorized signal is further denoted as

$$\widetilde{\mathbf{y}} = \text{vec}\{\mathbf{Y}_{\text{r}}\} = \mathbf{y} + \mathbf{n}, \tag{22}$$

where the definitions of $\mathbf{y} \triangleq \alpha_t \text{vec}\{\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{W}\mathbf{S}\}$ and $\mathbf{n} \triangleq \text{vec}\{\mathbf{B}^H \tilde{\mathbf{G}}_r^H \boldsymbol{\Theta} \mathbf{Z}_1 + \mathbf{N}_{\text{r}}\}$. From the perspective of radar sensing, we focus on parameter estimation performance in terms of CRB. We define the unknown target parameters as $\boldsymbol{\xi} \triangleq [\boldsymbol{\theta}^T, \boldsymbol{\alpha}^T]^T$ with $\boldsymbol{\theta} \triangleq [\theta_{\text{h}}, \theta_{\text{v}}]^T$ and $\boldsymbol{\alpha} \triangleq [\Re\{\alpha_t\}, \Im\{\alpha_t\}]^T$. Based on the complex observation $\widetilde{\mathbf{y}} \sim \mathcal{CN}(\mathbf{y}, \mathbf{R}_n)$ with $\mathbf{R}_n \triangleq \mathbf{I}_T \otimes \tilde{\mathbf{R}}_n$, where $\tilde{\mathbf{R}}_n \triangleq \sigma_{\text{z}}^2 \mathbf{B}^H \tilde{\mathbf{G}}_r^H \boldsymbol{\Theta}\boldsymbol{\Theta}^H \tilde{\mathbf{G}}_r \mathbf{B} + \sigma_{\text{r}}^2 \mathbf{I}_N$. According to the definition of the definitions of $\mathbf{y} \triangleq \alpha_t \text{vec}\{\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{W}\mathbf{S}\}$, the derivatives of $\mathbf{y}$ with respect to each parameter is given by

$$\frac{\partial \mathbf{y}}{\partial \theta_{\text{h}}} = \alpha_t \text{vec}\{\dot{\mathbf{F}}_{\theta_{\text{h}}}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{W}\mathbf{S}\}, \tag{23}$$

$$\frac{\partial \mathbf{y}}{\partial \theta_{\text{v}}} = \alpha_t \text{vec}\{\dot{\mathbf{F}}_{\theta_{\text{v}}}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{W}\mathbf{S}\}, \tag{24}$$

$$\frac{\partial \mathbf{y}}{\partial \boldsymbol{\alpha}} = [1, j] \otimes \text{vec}\{\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{W}\mathbf{S}\}, \tag{25}$$

where $\dot{\mathbf{F}}_{\theta_{\text{h}}}(\theta_{\text{h}}, \theta_{\text{v}}) \triangleq \frac{\partial \mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})}{\partial \theta_{\text{h}}}$ and $\dot{\mathbf{F}}_{\theta_{\text{v}}}(\theta_{\text{h}}, \theta_{\text{v}}) \triangleq \frac{\partial \mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})}{\partial \theta_{\text{v}}}$. We denote $\mathbf{M}$ as the Fisher information matrix (FIM), each element of FIM can be calculated as

$$\mathbf{M}(i,j) = \text{Tr}\left\{\mathbf{R}_{\text{n}}^{-1} \frac{\partial \mathbf{R}_{\text{n}}}{\partial \boldsymbol{\xi}_i} \mathbf{R}_{\text{n}}^{-1} \frac{\partial \mathbf{R}_{\text{n}}}{\partial \boldsymbol{\xi}_j}\right\} + 2\Re\left\{\frac{\partial \mathbf{y}^H}{\partial \boldsymbol{\xi}_i} \mathbf{R}_{\text{n}}^{-1} \frac{\partial \mathbf{y}}{\partial \boldsymbol{\xi}_j}\right\}. \tag{26}$$

To facilitate the derivation, the FIM can be divided into 4 blocks as

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\theta}^T} & \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T} \\ \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T}^T & \mathbf{M}_{\boldsymbol{\alpha}\boldsymbol{\alpha}^T} \end{bmatrix} = \begin{bmatrix} \mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}^T} & \mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T} \\ \mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T}^T & \mathbf{C}_{\boldsymbol{\alpha}\boldsymbol{\alpha}^T} \end{bmatrix}^{-1}, \tag{27}$$

where

$$\mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\theta}^T} = \begin{bmatrix} M_{\theta_{\text{h}}\theta_{\text{h}}}, M_{\theta_{\text{h}}\theta_{\text{v}}} \\ M_{\theta_{\text{v}}\theta_{\text{h}}}, M_{\theta_{\text{v}}\theta_{\text{v}}} \end{bmatrix}, \tag{28}$$

$$\mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T} = 2T\Re\left\{\begin{bmatrix} \alpha_t^* \text{Tr}\left(\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{R}\dot{\mathbf{F}}_{\theta_{\text{h}}}^H(\theta_{\text{h}}, \theta_{\text{v}})\tilde{\mathbf{R}}_n^{-1}\right) \\ \alpha_t^* \text{Tr}\left(\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{R}\dot{\mathbf{F}}_{\theta_{\text{v}}}^H(\theta_{\text{h}}, \theta_{\text{v}})\tilde{\mathbf{R}}_n^{-1}\right) \end{bmatrix} [1, j]\right\}, \tag{29}$$

and

$$\mathbf{M}_{\boldsymbol{\alpha}\boldsymbol{\alpha}^T} = 2T\text{Tr}(\mathbf{F}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{R}\mathbf{F}^H(\theta_{\text{h}}, \theta_{\text{v}})\tilde{\mathbf{R}}_n^{-1})\mathbf{I}_2, \tag{30}$$

where

$$M_{\theta_{\text{s}}\theta_{\text{z}}} \triangleq 2T|\alpha_t|^2 \Re\{\text{Tr}(\dot{\mathbf{F}}_{\theta_{\text{s}}}(\theta_{\text{h}}, \theta_{\text{v}})\mathbf{R}\dot{\mathbf{F}}_{\theta_{\text{z}}}^H(\theta_{\text{h}}, \theta_{\text{v}})\tilde{\mathbf{R}}_n^{-1})\} \tag{31}$$

for $\theta_{\text{s}} \in \{\theta_{\text{h}}, \theta_{\text{v}}\}$ and $\theta_{\text{z}} \in \{\theta_{\text{h}}, \theta_{\text{v}}\}$. Thus, the CRB for estimating $\boldsymbol{\theta}$ can be expressed as

$$\begin{aligned} \text{CRB}_{\boldsymbol{\theta}} &= \text{CRB}_{\theta_{\text{h}}} + \text{CRB}_{\theta_{\text{v}}} = \text{Tr}(\mathbf{C}_{\boldsymbol{\theta}\boldsymbol{\theta}^T}) \\ &= \text{Tr}\left((\mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\theta}^T} - \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T}\mathbf{M}_{\boldsymbol{\alpha}\boldsymbol{\alpha}^T}^{-1}\mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T}^T)^{-1}\right). \end{aligned} \tag{32}$$

D. Problem Formulation

In this paper, we aim to maximize the minimum SR by jointly designing transmit beamforming $\mathbf{W}$, active RIS phase shift matrix $\boldsymbol{\Theta}$ and the transmit and receive MA position selection matrix $\mathbf{A}$ and $\mathbf{B}$ while assuring the QoS requirements of communication users. The SR optimization problem can be formulated as

$$\max_{\mathbf{W},\boldsymbol{\Theta},\mathbf{A},\mathbf{B}} \quad \min_{k\in\mathcal{K}} \quad R_{\text{sec},k} \tag{33a}$$

$$\text{s.t. } \gamma_{\text{c},k} \geq \Gamma, \forall k \in \mathcal{K}, \tag{33b}$$

$$\text{CRB}_{\boldsymbol{\theta}} \leq \delta, \tag{33c}$$

$$\|\mathbf{W}\|_F^2 \leq P_{\text{BS}}, \tag{33d}$$

$$P(\mathbf{W}, \boldsymbol{\Theta}) \leq P_{\text{RIS}}, \tag{33e}$$

$$0 \leq \beta_l \leq \beta_{\max}, \forall l \in \mathcal{L}, \tag{33f}$$

$$a_{m,i} \in \{0,1\}, \forall m \in \mathcal{M}, \forall i \in \mathcal{I}, \tag{33g}$$

$$\sum_{i=1}^{I} a_{m,i} = 1, \forall m \in \mathcal{M}, \tag{33h}$$

$$\mathbf{a}_m^T \mathbf{D}_1 \mathbf{a}_{m'} \geq D_{1,\min}, 1 \leq m \neq m' \leq M, \tag{33i}$$

$$b_{n,j} \in \{0,1\}, \forall n \in \mathcal{N}, \forall j \in \mathcal{J}, \tag{33j}$$

$$\sum_{j=1}^{J} b_{n,j} = 1, \forall n \in \mathcal{N}, \tag{33k}$$

$$\mathbf{a}_n^T \mathbf{D}_2 \mathbf{a}_{n'} \geq D_{2,\min}, 1 \leq n \neq n' \leq N, \tag{33l}$$

where $\mathbf{D}_1$ and $\mathbf{D}_2$ are the distance matrix at transmit and receive MA, respectively. The distance between the centres of any two transmitter and receiver MA elements must exceed the distance thresholds $D_{1,\min}$ and $D_{2,\min}$, respectively. In addition, $P(\mathbf{W}, \boldsymbol{\Theta})$ is the power consumption at the active RIS and can be expressed as follows

$$\begin{aligned} P(\mathbf{W}, \boldsymbol{\Theta}) = &\|\boldsymbol{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\|_F^2 + \sigma_{\text{t}}^2\|\boldsymbol{\Theta}^H\boldsymbol{\Lambda}(\theta_{\text{h}}, \theta_{\text{v}})\boldsymbol{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\|_F^2 \\ &+ \sigma_{\text{t}}^2\sigma_{\text{z}}^2\|\boldsymbol{\Theta}^H\boldsymbol{\Lambda}(\theta_{\text{h}}, \theta_{\text{v}})\boldsymbol{\Theta}\|_F^2 + 2\sigma_{\text{z}}^2\|\boldsymbol{\Theta}\|_F^2. \end{aligned} \tag{34}$$

It can be noted that the constraint (33b) ensures the QoS requirements of communication users. (33d) and (33e) represent the power constraints, where P_{BS} and P_{RIS} are the maximum power threshold at the BS and active RIS, respectively. (33f) is the maximum amplitude constraint of the active RIS. (33g), (33h), and (33i) ensures compliance with the selection requirements for the MA position and the distance constraints between elements. (33j), (33k), and (33l) also have the same selection requirements for receiving MA antennas.

Due to antenna mobility and channel variability, the environment is highly dynamic, posing challenges for conventional optimization methods. Moreover, the original optimization problem cannot directly handle constraints (33i) and (33l) using traditional optimization methods, and directly using DRL algorithms will result in extremely slow convergence and the ultimate outcome cannot be guaranteed. Therefore, we extracts

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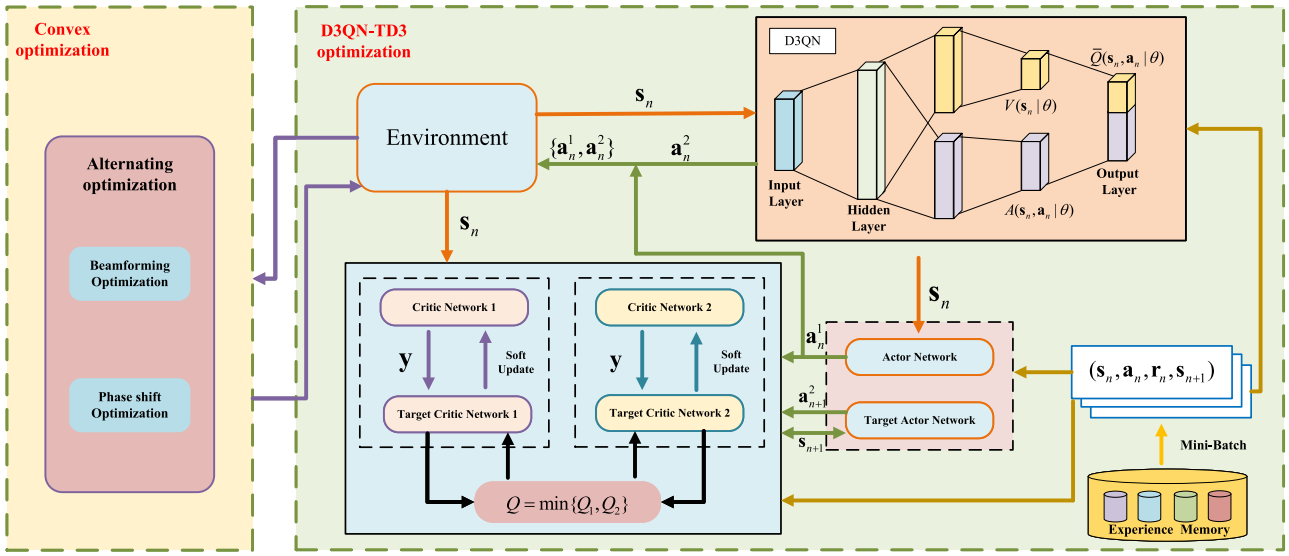

Fig. 2. Flowchart of the AO-based D3QN-TD3 method.

an optimization subproblem $\mathbf{P}_1$ concerning the optimization variable $\mathbf{W}$ and $\mathbf{\Theta}$ from the overall optimization problem

$$\mathbf{P}_1: \max_{\mathbf{W},\mathbf{\Theta}} \min_{k\in\mathcal{K}} \quad R_{\text{sec},k} \tag{35a}$$

$$\text{s.t.} \quad (33b)-(33f). \tag{35b}$$

In addition, we employs a novel hybrid method combining AO-based approach and DRL technology to tackle complex challenges in Problem (33).

IV. Proposed AO-Based D3QN-TD3 Method

A. AO-Based DRL Structure

Due to the particularity of MA position optimization, traditional optimization methods are difficult to solve the MA optimization of discrete positions. Therefore, in order to obtain better MA selection, we adopt DRL for antenna optimization. Considering that DRL may be difficult to converge to the optimal solution in continuous space optimization, we adopt an AO algorithm to optimize beamforming and phase parameters, thereby effectively improving the convergence efficiency towards the optimal solution. The flowchart of the AO-based D3QN-TD3 method is shown in Fig. 2.

1) MDP Model: Given the complex and dynamic nature of the proposed MA-enabled active RIS-aided ISAC system, a DRL method was adopted to address this challenge. By formalizing the optimization problem of transmission rate as a Markov decision process, the complexity of the optimization problem was effectively simplified. Within this framework, the semantic communication network is enhanced through the collaborative interaction between BS and RIS, creating a dynamic environment. The BS control center acts as an intelligent agent responsible for decision-making and regulating the system's behavior. The problem involves various elements such as the environment, intelligent agents, state space, action space, state transition probabilities, and reward functions. These elements are crucial for accurately modeling and solving the Markov decision process, as detailed below.

- *State space* $\mathcal{S}$: The state of the system at time slot n is defined as

$$\boldsymbol{s}_n = \{\{R_{\text{sec},k}\}_n, \{\text{CRB}_{\boldsymbol{\theta}}\}_n, \{\mathbf{G}_t\}_n, \{\mathbf{G}_r\}_n\}, \tag{36}$$

where $\{R_{\text{sec},k}\}_n$ is the security rate of Bob k, $\{\text{CRB}_{\boldsymbol{\theta}}\}_n$ is the estimated CRB, $\{\mathbf{G}_t\}_n$ and $\{\mathbf{G}_r\}_n$ are the channel matrices of the BS-RIS transmitting and RIS-BS receiving paths, respectively. $\boldsymbol{s}_n \in \mathcal{S}$, $\mathcal{S}$ is the set of all state space.

- Action space $\mathcal{A}$: Due to the fact that optimization problems involve a mixture of discrete and continuous actions, the action space is divided into two parts. One part is used for discrete actions, including the transmit and receive MA position selection matrices $\mathbf{A}_n$ and $\mathbf{B}_n$, and the other part is used for continuous actions, including the beamforming vector $\mathbf{W}_n$ of the base station and the phase shift $\mathbf{\Theta}_n$ of the active RIS. The state of the system at time slot n is defined as

$$\boldsymbol{a}_n = \{\mathbf{W}_n, \mathbf{\Theta}_n, \mathbf{A}_n, \mathbf{B}_n\}, \tag{37}$$

where $\mathbf{W}_n$ is the transmit beamforming matrix, $\mathbf{\Theta}_n$ is the active RIS phase shift matrix and $\mathbf{A}_n$ and $\mathbf{B}_n$ are the transmit and receive MA position selection matrices, respectively. $\boldsymbol{a}_n \in \mathcal{A}$, $\mathcal{A}$ is the set of all state space.

- Reward function $\mathcal{R}$: The reward of the system at time slot n is defined by $\boldsymbol{r}_n$, which is based on MAs' selection of antenna location and quantified information feedback from the environment, and can be calculated as

$$\boldsymbol{r}_n = R + \lambda_1 \sum_{k=1}^{K} r_{\text{c},k} + \lambda_2 r_{\text{r}}, \tag{38}$$

where λ_1 and λ_2 are constant coefficients, setting the sum SR, each user achievable rate, and radar estimation error

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within reasonable numerical ranges. $R = \sum_{k=1}^{K} R_{\text{sec},k}$ denotes the sum SR, and each user communication requirement and CRB estimation are given respectively as

$$r_{\text{c},k} = \begin{cases} \gamma_{\text{c},k} - \Gamma, & \text{if } \gamma_{\text{c},k} > \Gamma, \\ 0, & \text{otherwise}, \end{cases} \tag{39}$$

$$r_{\text{r}} = \begin{cases} (\delta - \text{CRB}_{\boldsymbol{\theta}})/0.05, & \text{if } \text{CRB}_{\boldsymbol{\theta}} < \delta, \\ 0, & \text{otherwise}, \end{cases} \tag{40}$$

The goal of MDP is to maximize the long-term accumulative reward, which can be expressed as

$$R_n = \sum_{i=0}^{n} \gamma^i \boldsymbol{r}_i. \tag{41}$$

When the optimal robust SR performance is achieved, the system will continue to maintain the state to maximize the cumulative expected reward. Therefore, by utilizing the feature of maximizing the cumulative expectation of rewards in MDP, MA optimization will converge to the optimal selection position and remain optimal at this position.

B. Proposed AO-Based Algorithm

1) Problem Reformulation: To begin with, we notice that the max-min objective function (35a) is non-smooth, hence, an auxiliary variable ξ is introduced. Moreover, an auxiliary variable $\mathbf{J} \in \mathbb{C}^{2\times 2}$ is introduced to handle the CRB constraint, and the constraint (33c) can be converted into the following two constraints:

$$\text{Tr}(\mathbf{J}^{-1}) \leq \delta, \tag{42a}$$

$$\begin{bmatrix} \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\theta}^T} - \mathbf{J} & \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T} \\ \mathbf{M}_{\boldsymbol{\theta}\boldsymbol{\alpha}^T}^T & \mathbf{M}_{\boldsymbol{\alpha}\boldsymbol{\alpha}^T} \end{bmatrix} \succeq \mathbf{0}. \tag{42b}$$

Correspondingly, the original Problem $\mathbf{P}_1$ is expressed as

$$\max_{\mathbf{W},\boldsymbol{\Theta},\mathbf{J},\xi} \quad \xi \tag{43a}$$

$$\text{s.t.} \quad R_{\text{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{43b}$$

$$(33\text{b}), (42\text{a})-(42\text{b}), (33\text{d})-(33\text{f}), \tag{43c}$$

where ξ is an auxiliary variable to ensure the equivalence between constraint (43b) and the objective function of the original optimization problem. Notably, Problem (43) aimed at maximizing the long-term sum SR, which is non-convex and characterized by numerous time-varying optimization variables.

2) Optimization of $\mathbf{J}$*:* Given other variables, the optimization problem for updating $\mathbf{J}$ can be formulated as

$$\max_{\mathbf{J},\xi} \quad \xi \tag{44a}$$

$$\text{s.t.} \quad (42\text{a}) - (42\text{b}). \tag{44b}$$

We observe that Problem (44) is a semidefinite programming (SDP) problem, which can be readily solved by standard algorithms.

3) Optimization of $\mathbf{W}$*:* To handle the SINR constraint (33b), the SINR constraint of k-th user is reformed as

$$(1+\Gamma^{-1})\text{Tr}(\mathbf{R}_k\mathbf{U}_k) \geq \text{Tr}(\mathbf{R}\mathbf{U}_k) + \|\mathbf{h}_{\text{r},k}^H\boldsymbol{\Theta}\|^2\sigma_{\text{z}}^2 + \sigma_k^2, \forall k \in \mathcal{K}, \tag{45}$$

where $\mathbf{U}_k \triangleq \mathbf{A}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{A}$. To facilitate the handling of constraint (43b), we reformulate the achievable rate of Bob and Eve in (43b) as follows, respectively

$$R_{\text{c},k} = \log_2\left(1+\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_k\mathbf{R}_i)\right) - \log_2\left(1+\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_k\mathbf{R}_i)\right), \tag{46}$$

$$R_{\text{e},k} = \log_2\left(1+\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_{\text{r}}\mathbf{R}_i)\right) - \log_2\left(1+\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_{\text{r}}\mathbf{R}_i)\right), \tag{47}$$

where $\mathbf{H}_k = \frac{\mathbf{A}^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{A}}{\|\mathbf{h}_{\text{r},k}^H\boldsymbol{\Theta}\|^2\sigma_z^2+\sigma_k^2}$ and $\mathbf{H}_{\text{r}} = \frac{\mathbf{A}^H\mathbf{h}_{\text{r}}\mathbf{h}_{\text{r}}^H\mathbf{A}}{\|\mathbf{h}_{\text{r},t}^H\boldsymbol{\Theta}\|^2\sigma_z^2+\sigma_{\text{e}}^2}$. Consequently, Problem (43) can be reduced as

$$\max_{\mathbf{R},\{\mathbf{R}_k\},\xi} \quad \xi \tag{48a}$$

$$\text{s.t.} \quad R_{\text{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{48b}$$

$$\text{Tr}(\mathbf{R}) \leq P_{\text{BS}}, \mathbf{R} \succeq \mathbf{0}, \tag{48c}$$

$$\text{Tr}(\mathbf{R}\mathbf{Y}_1) + \sigma_{\text{t}}^2\text{Tr}(\mathbf{R}\mathbf{Y}_2) \leq \tilde{P}, \tag{48d}$$

$$\mathbf{R}_k \succeq \mathbf{0}, \forall k \in \mathcal{K}, \tag{48e}$$

$$\mathbf{R} - \sum_{k=1}^{K}\mathbf{R}_k \succeq \mathbf{0}, \tag{48f}$$

$$\text{rank}(\mathbf{R}_k) = 1, \tag{48g}$$

$$(42\text{b}), (45), \tag{48h}$$

where $\mathbf{Y}_1 \triangleq \mathbf{A}^H\tilde{\mathbf{G}}_t^H\boldsymbol{\Theta}^H\boldsymbol{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}$, $\mathbf{Y}_2 \triangleq \mathbf{A}^H\tilde{\mathbf{G}}_t^H\boldsymbol{\Theta}^H\boldsymbol{\Lambda}^H(\theta_{\text{h}}, \theta_{\text{v}})\boldsymbol{\Theta}\boldsymbol{\Theta}^H\boldsymbol{\Lambda}(\theta_{\text{h}}, \theta_{\text{v}})\boldsymbol{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}$ and $\tilde{P} \triangleq P_{\text{RIS}} - \sigma_{\text{t}}^2\sigma_{\text{z}}^2\|\boldsymbol{\Theta}^H\boldsymbol{\Lambda}(\theta_{\text{h}}, \theta_{\text{v}})\boldsymbol{\Theta}\|_F^2 - 2\sigma_{\text{z}}^2\|\boldsymbol{\Theta}\|_F^2$. Then, we apply SCA method to deal with the non-convex rank-one constraint (48b)

$$R_{\text{c},k} \geq \log_2\left(1+\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_k\mathbf{R}_i)\right) - \log_2\left(1+\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_k\mathbf{R}_i^{(j)})\right) - \frac{\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_k(\mathbf{R}_i - \mathbf{R}_i^{(j)}))}{\ln 2(1+\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_k\mathbf{R}_i^{(j)}))} \triangleq \hat{R}_{\text{c},k}, \tag{49}$$

$$R_{\text{e},k} \leq \log_2\left(1+\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_{\text{r}}\mathbf{R}_i^{(j)})\right) - \log_2\left(1+\sum_{i\neq k}^{K}\text{Tr}(\mathbf{H}_{\text{r}}\mathbf{R}_i)\right) + \frac{\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_{\text{r}}(\mathbf{R}_i - \mathbf{R}_i^{(j)}))}{\ln 2\left(1+\sum_{i=1}^{K}\text{Tr}(\mathbf{H}_{\text{r}}\mathbf{R}_i^{(j)})\right)} \triangleq \hat{R}_{\text{e},k}, \tag{50}$$

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where $\mathbf{R}_i^{(j)}$ denotes the optimal $\mathbf{R}_i$ in the j-iteration. Thus, the left-hand of (52b) is bounded below as

$$R_{\mathrm{c},k} - R_{\mathrm{e},k} \geq \hat{R}_{\mathrm{c},k} - \hat{R}_{\mathrm{e},k} \triangleq \hat{R}_{\mathrm{sec},k}. \tag{51}$$

The final optimization problem can be expressed as

$$\max_{\mathbf{R},\{\mathbf{R}_k\},\xi} \quad \xi \tag{52a}$$

$$\text{s.t.} \quad \hat{R}_{\mathrm{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{52b}$$

$$(42\text{b}), (48\text{c}) - (48\text{g}), (45), \tag{52c}$$

where ρ_k is the penalty factor. It is clear that omitting (48f) reformulates the problem as a semidefinite programming problem. To recover the optimal rank-one solution, we convert the rank-one constraint into an alternative form. Thus, we reformulate the original Problem (48) by incorporating the penalty terms $\mathrm{Tr}(\mathbf{R}_k) - \lambda_{\max}(\mathbf{R}_k)$ into the objective function, and the first-order Taylor approximation is constructed to tackle the objective function

$$\lambda_{\max}(\mathbf{R}_k) \geq \lambda_{\max}(\mathbf{R}_k^{(j)}) + \mathrm{Tr}\left(\boldsymbol{\mu}(\mathbf{R}_k^{(j)})\boldsymbol{\mu}^H(\mathbf{R}_k^{(j)})(\mathbf{R}_k - \mathbf{R}_k^{(j)})\right), \tag{53}$$

where $\boldsymbol{\mu}(\mathbf{R}_k^{(j)})$ denotes the unit eigenvector associated with the maximum eigenvalue of $\mathbf{R}_k^{(j)}$. Thus, the Problem (52) can be formulated as

$$\max_{\mathbf{R},\{\mathbf{R}_k\},\xi} \quad \xi - \sum_{k=1}^{K} \rho_k \Big[\mathrm{Tr}(\mathbf{R}_k) - \lambda_{\max}(\mathbf{R}_k^{(j)}) - \mathrm{Tr}\left(\boldsymbol{\mu}(\mathbf{R}_k^{(j)})\boldsymbol{\mu}^H(\mathbf{R}_k^{(j)})(\mathbf{R}_k - \mathbf{R}_k^{(j)})\right)\Big] \tag{54a}$$

$$\text{s.t.} \quad \hat{R}_{\mathrm{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{54b}$$

$$(42\text{a}) - (42\text{b}), (48\text{c}) - (48\text{f}), (45). \tag{54c}$$

It noted that problem (54) belongs to the class of convex SDP problems, characterized by linear or semidefinite constraints. The optimal solution can be derived efficiently with convex optimization toolbox i.e., CVX. Denoting $\tilde{\mathbf{R}}, \{\tilde{\mathbf{R}}_k\}$ as a feasible solution to the Problem (48), the lemma below is introduced to construct the corresponding rank-one solution $\hat{\mathbf{R}}$, $\{\hat{\mathbf{R}}_k\}$ to the Problem (54). Therefore, we can obtain a solution of $\mathbf{W}$ for (54) as $\hat{\mathbf{W}} = [\hat{\mathbf{W}}_{\mathrm{c}}, \hat{\mathbf{W}}_{\mathrm{r}}]$.

Lemma 1: By omitting the rank-one constraint, the optimal communication beamforming matrix $\hat{\mathbf{W}}_{\mathrm{c}} = [\hat{\mathbf{w}}_{\mathrm{c},1}, \hat{\mathbf{w}}_{\mathrm{c},2}, \ldots, \hat{\mathbf{w}}_{\mathrm{c},K}]$ to the Problem (54) can be constructed as

$$\hat{\mathbf{w}}_{\mathrm{c},k} = \frac{\tilde{\mathbf{R}}_k \mathbf{A}^H \mathbf{h}_k}{\sqrt{\mathbf{h}_k^H \mathbf{A} \tilde{\mathbf{R}}_k \mathbf{A}^H \mathbf{h}_k}}, \hat{\mathbf{R}}_k = \hat{\mathbf{w}}_{\mathrm{c},k}\hat{\mathbf{w}}_{\mathrm{c},k}^H, \forall k \in \mathcal{K}. \tag{55}$$

According to Cholesky decomposition, the radar beamforming matrix $\hat{\mathbf{W}}_{\mathrm{r}} = [\hat{\mathbf{w}}_{\mathrm{r},1}, \hat{\mathbf{w}}_{\mathrm{r},2}, \ldots, \hat{\mathbf{w}}_{\mathrm{r},M}]$ is obtained as

$$\hat{\mathbf{W}}_{\mathrm{r}}\hat{\mathbf{W}}_{\mathrm{r}}^H = \tilde{\mathbf{R}} - \sum_{k=1}^{K} \hat{\mathbf{R}}_k. \tag{56}$$

Proof: See Reference [85].

4) Optimization of $\boldsymbol{\Theta}$*:* When fixed the other variables, the optimization problem related to $\boldsymbol{\Theta}$ can be formulated as

$$\max_{\boldsymbol{\Theta},\xi} \quad \xi \tag{57a}$$

$$\text{s.t.} \quad R_{\mathrm{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{57b}$$

$$(42\text{b}), (33\text{b}), (33\text{e}) - (33\text{f}). \tag{57c}$$

Due to the difficulty in handling the quartic variable within the constraints, we define $\mathbf{v} = [1, \boldsymbol{\theta}^H]^H$, $\mathbf{V} = \mathbf{v}\mathbf{v}^H$, $\mathrm{rank}(\mathbf{V}) = 1$. Firstly, we deal with the constraint (57b) by introducing the following matrices

$$\mathbf{L}_k \triangleq \frac{\sigma_{\mathrm{z}}^2}{\sigma_k^2}\begin{bmatrix} 0 & \mathbf{0}_{1\times L} \\ \mathbf{0}_{L\times 1} & \mathrm{diag}(\mathbf{h}_{\mathrm{r},k})\mathrm{diag}(\mathbf{h}_{\mathrm{r},k}^H) \end{bmatrix}, \tag{58}$$

$$\mathbf{U}_k \triangleq \frac{1}{\sigma_k^2}\begin{bmatrix} \mathbf{h}_{\mathrm{d},k}^H \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},k}^H)\tilde{\mathbf{G}}_t \end{bmatrix} \sum_{i=1}^{K} \mathbf{R}_i \begin{bmatrix} \mathbf{h}_{\mathrm{d},k}^H \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},k}^H)\tilde{\mathbf{G}}_t \end{bmatrix}^H + \mathbf{L}_k, \tag{59}$$

$$\mathbf{Y}_{\mathrm{c},k} \triangleq \frac{1}{\sigma_k^2}\begin{bmatrix} \mathbf{h}_{\mathrm{d},k}^H \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},k}^H)\tilde{\mathbf{G}}_t \end{bmatrix} \sum_{i\neq k}^{K} \mathbf{R}_i \begin{bmatrix} \mathbf{h}_{\mathrm{d},k}^H \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},k}^H)\tilde{\mathbf{G}}_t \end{bmatrix}^H + \mathbf{L}_k, \tag{60}$$

$$\mathbf{L}_{\mathrm{e}} \triangleq \frac{\sigma_{\mathrm{z}}^2}{\sigma_k^2}\begin{bmatrix} 0 & \mathbf{0}_{1\times L} \\ \mathbf{0}_{L\times 1} & \mathrm{diag}(\mathbf{h}_{\mathrm{r},t})\mathrm{diag}(\mathbf{h}_{\mathrm{r},t}^H) \end{bmatrix}, \tag{61}$$

$$\mathbf{U}_{\mathrm{e}} \triangleq \frac{1}{\sigma_k^2}\begin{bmatrix} \mathbf{0}_{1\times M} \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},t}^H)\tilde{\mathbf{G}}_t \end{bmatrix} \sum_{i=1}^{K} \mathbf{R}_i \begin{bmatrix} \mathbf{0}_{1\times M} \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},t}^H)\tilde{\mathbf{G}}_t \end{bmatrix}^H + \mathbf{L}_e, \tag{62}$$

$$\mathbf{Y}_{\mathrm{e},k} \triangleq \frac{1}{\sigma_k^2}\begin{bmatrix} \mathbf{0}_{1\times M} \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},t}^H)\tilde{\mathbf{G}}_t \end{bmatrix} \sum_{i\neq k}^{K} \mathbf{R}_i \begin{bmatrix} \mathbf{0}_{1\times M} \\ \mathrm{diag}(\mathbf{h}_{\mathrm{r},t}^H)\tilde{\mathbf{G}}_t \end{bmatrix}^H + \mathbf{L}_e. \tag{63}$$

Thus, the achievable rate of Bob k and Eve are expressed by $R_{\mathrm{c},k} = \log_2(1 + \mathrm{Tr}(\mathbf{U}_k\mathbf{V})) - \log_2(1 + \mathrm{Tr}(\mathbf{Y}_{\mathrm{c},k}\mathbf{V}))$ and $R_{\mathrm{e},k} = \log_2(1 + \mathrm{Tr}(\mathbf{U}_{\mathrm{e}}\mathbf{V})) - \log_2(1 + \mathrm{Tr}(\mathbf{Y}_{\mathrm{e},k}\mathbf{V}))$, respectively. Similarly, we apply SCA method to deal with the non-convex rank-one constraint (57b), can be obtained as

$$R_{\mathrm{c},k} \geq \log_2(1 + \mathrm{Tr}(\mathbf{U}_k\mathbf{V})) - \log_2(1 + \mathrm{Tr}(\mathbf{Y}_{\mathrm{c},k}\mathbf{V}^{(j)})) - \frac{\mathrm{Tr}((\mathbf{Y}_{\mathrm{c},k}(\mathbf{V} - \mathbf{V}^{(j)}))}{\ln 2(1 + \mathrm{Tr}((\mathbf{Y}_{\mathrm{c},k}\mathbf{V}^{(j)}))} \triangleq \tilde{R}_{\mathrm{c},k}, \tag{64}$$

$$R_{\mathrm{e},k} \leq \log_2(1 + \mathrm{Tr}(\mathbf{U}_{\mathrm{e}}\mathbf{V}^{(j)})) - \log_2(1 + \mathrm{Tr}(\mathbf{Y}_{\mathrm{e},k}\mathbf{V})) + \frac{\mathrm{Tr}(\mathbf{U}_{\mathrm{e}}(\mathbf{V} - \mathbf{V}^{(j)}))}{\ln 2(1 + \mathrm{Tr}(\mathbf{U}_{\mathrm{e}}\mathbf{V}^{(j)}))} \triangleq \tilde{R}_{\mathrm{e},k}, \tag{65}$$

where $\mathbf{V}^{(j)}$ denotes the optimal $\mathbf{V}$ in the j-iteration. Thus, the left-hand of (57b) is bounded below as

$$R_{\mathrm{c},k} - R_{\mathrm{e},k} \geq \tilde{R}_{\mathrm{c},k} - \tilde{R}_{\mathrm{e},k} \triangleq \tilde{R}_{\mathrm{sec},k}. \tag{66}$$

Note that for the diagonal matrix, we have $\hat{\mathbf{V}} \triangleq \boldsymbol{\Theta}\boldsymbol{\Theta}^{\mathrm{H}} = \mathrm{Diag}(\mathrm{diag}([\mathbf{V}]_{2:L+1,2:L+1}))$. Then, the SINR constraint of (33b)

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can be reformed as

$$(1+\Gamma^{-1})\mathrm{Tr}(\mathbf{R}_{1,k}\mathbf{V}) - \mathrm{Tr}(\mathbf{R}_{2,k}\mathbf{V}) \geq \sigma_{\mathrm{z}}^2\mathrm{Tr}(\hat{\mathbf{V}}\mathbf{h}_{\mathrm{r},k}\mathbf{h}_{\mathrm{r},k}^H) + \sigma_k^2, \forall k \in \mathcal{K}. \tag{67}$$

where, $\mathbf{R}_{1,k} \triangleq \mathbf{H}_k\mathbf{R}_k\mathbf{H}_k^H$, $\mathbf{R}_{2,k} \triangleq \mathbf{H}_k\mathbf{R}\mathbf{H}_k^H$, and $\mathbf{H}_k \triangleq [\mathbf{G}^H\mathrm{Diag}(\mathbf{h}_{1,k}), \mathbf{h}_{2,k}]^H$.

Next, we define $\mathbf{F}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}) \triangleq \mathbf{M}_1\mathbf{V}\mathbf{M}_2$, $\dot{\mathbf{F}}_{\theta_{\mathrm{h}}}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}) \triangleq \mathbf{M}_3\mathbf{V}\mathbf{M}_4$ and $\dot{\mathbf{F}}_{\theta_{\mathrm{v}}}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}) \triangleq \mathbf{M}_5\mathbf{V}\mathbf{M}_6$, where

$$\mathbf{M}_1 \triangleq \left[\mathbf{0}, \mathbf{B}^H\tilde{\mathbf{G}}_r^H\mathrm{diag}(\mathbf{a}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}))\right], \tag{68a}$$

$$\mathbf{M}_2^H \triangleq \left[\mathbf{0}, \mathrm{diag}(\mathbf{a}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}))\tilde{\mathbf{G}}_t\mathbf{A}\right], \tag{68b}$$

$$\mathbf{M}_3 \triangleq \left[\mathbf{0}, \mathbf{B}^H\tilde{\mathbf{G}}_r^H\mathrm{diag}(\dot{\mathbf{a}}_{\theta_{\mathrm{h}}}(\theta_{\mathrm{h}},\theta_{\mathrm{h}})\right], \tag{68c}$$

$$\mathbf{M}_4^H \triangleq \left[\mathbf{0}, \mathrm{diag}(\dot{\mathbf{a}}_{\theta_{\mathrm{v}}}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}}))\tilde{\mathbf{G}}_t\mathbf{A}\right], \tag{68d}$$

$$\mathbf{M}_5 \triangleq \left[\mathbf{0}, \mathbf{B}^H\tilde{\mathbf{G}}_r^H\mathrm{diag}(\dot{\mathbf{a}}_{\theta_{\mathrm{v}}}(\theta_{\mathrm{h}},\theta_{\mathrm{v}})\right], \tag{68e}$$

$$\mathbf{M}_6^H \triangleq \left[\mathbf{0}, \mathrm{diag}(\dot{\mathbf{a}}_{\theta_{\mathrm{v}}}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}}))\tilde{\mathbf{G}}_t\mathbf{A}\right], \tag{68f}$$

and where $\dot{\mathbf{a}}_{\theta_{\mathrm{h}}}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}) \triangleq \frac{\partial\mathbf{a}(\theta_{\mathrm{h}},\theta_{\mathrm{v}})}{\partial\theta_{\mathrm{h}}}$ and $\dot{\mathbf{a}}_{\theta_{\mathrm{v}}}(\theta_{\mathrm{h}},\theta_{\mathrm{v}}) \triangleq \frac{\partial\mathbf{a}(\theta_{\mathrm{h}},\theta_{\mathrm{v}})}{\partial\theta_{\mathrm{v}}}$. Next, we introduce an auxiliary variable $\mathbf{f} \triangleq [f_1, f_2, f_3, f_4, f_5, f_6]^T$ to take $\mathbf{V}$ out of the positive semidefinite matrix constraint

$$\begin{bmatrix} 2T|\alpha_t|^2\Re\left\{\begin{bmatrix} f_1 & f_2 \\ f_2 & f_4\end{bmatrix}\right\} - \mathbf{J} & 2T\Re\left\{\alpha_{\mathrm{t}}^*\begin{bmatrix} f_3 \\ f_5\end{bmatrix}[1j]\right\} \\ 2T\Re\left\{\alpha_{\mathrm{t}}^*\begin{bmatrix} 1 \\ j\end{bmatrix}[f_3 f_5]\right\} & 2Tf_6\mathbf{I}_2 \end{bmatrix} \succeq \mathbf{0}, \tag{69}$$

by introduce an auxiliary variable $\mathbf{V}_1 = \mathbf{V}$, the element of matrix can be expressed

$$f_1 = \mathrm{Tr}(\mathbf{M}_3\mathbf{V}\mathbf{M}_4\mathbf{R}\mathbf{M}_4^H\mathbf{V}^H\mathbf{M}_3^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{4,4}\mathbf{V}_1\mathbf{M}_{3,3}), \tag{70a}$$

$$f_2 = \mathrm{Tr}(\mathbf{M}_3\mathbf{V}\mathbf{M}_4\mathbf{R}\mathbf{M}_6^H\mathbf{V}^H\mathbf{M}_5^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{4,6}\mathbf{V}_1\mathbf{M}_{5,3}), \tag{70b}$$

$$f_3 = \mathrm{Tr}(\mathbf{M}_1\mathbf{V}\mathbf{M}_2\mathbf{R}\mathbf{M}_3^H\mathbf{V}^H\mathbf{M}_4^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{2,3}\mathbf{V}_1\mathbf{M}_{4,1}), \tag{70c}$$

$$f_4 = \mathrm{Tr}(\mathbf{M}_5\mathbf{V}\mathbf{M}_6\mathbf{R}\mathbf{M}_6^H\mathbf{V}^H\mathbf{M}_5^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{6,6}\mathbf{V}_1\mathbf{M}_{5,5}), \tag{70d}$$

$$f_5 = \mathrm{Tr}(\mathbf{M}_1\mathbf{V}\mathbf{M}_2\mathbf{R}\mathbf{M}_6^H\mathbf{V}^H\mathbf{M}_5^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{2,6}\mathbf{V}_1\mathbf{M}_{5,1}), \tag{70e}$$

$$f_6 = \mathrm{Tr}(\mathbf{M}_1\mathbf{V}\mathbf{M}_2\mathbf{R}\mathbf{M}_2^H\mathbf{V}^H\mathbf{M}_1^H\tilde{\mathbf{R}}_n^{-1}) \triangleq \mathrm{Tr}(\mathbf{V}\mathbf{M}_{2,2}\mathbf{V}_1\mathbf{M}_{1,1}), \tag{70f}$$

where $\mathbf{M}_{4,4} \triangleq \mathbf{M}_3^H\mathbf{R}\mathbf{M}_3$, $\tilde{\mathbf{M}}_{3,3} \triangleq \mathbf{M}_3^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_3$, $\mathbf{M}_{4,6} \triangleq \mathbf{M}_4^H\mathbf{R}\mathbf{M}_6$, $\mathbf{M}_{5,3} \triangleq \mathbf{M}_3^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_3$, $\mathbf{M}_{2,3} \triangleq \mathbf{M}_2^H\mathbf{R}\mathbf{M}_3$, $\mathbf{M}_{4,1} \triangleq \mathbf{M}_4^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_1$, $\mathbf{M}_{6,6} \triangleq \mathbf{M}_6^H\mathbf{R}\mathbf{M}_6$, $\mathbf{M}_{5,5} \triangleq \mathbf{M}_5^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_5$, $\mathbf{M}_{2,6} \triangleq \mathbf{M}_2^H\mathbf{R}\mathbf{M}_6$, $\mathbf{M}_{5,1} \triangleq \mathbf{M}_5^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_1$, $\mathbf{M}_{2,2} \triangleq \mathbf{M}_2^H\mathbf{R}\mathbf{M}_2$, and $\mathbf{M}_{1,1} \triangleq \mathbf{M}_1^H\tilde{\mathbf{R}}_n^{-1}\mathbf{M}_1$. And then we perform the following processing on constraint (33e), the fist term of (33e) can be expressed as

$$\mathrm{Tr}(\mathbf{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\mathbf{W}^H\mathbf{A}^H\tilde{\mathbf{G}}_t^H\mathbf{\Theta}^H) = \mathrm{Tr}(\mathbf{N}_1\hat{\mathbf{V}}), \tag{71}$$

where $\mathbf{N}_1 = \tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\mathbf{W}^H\mathbf{A}^H\tilde{\mathbf{G}}_t^H$. Then we define $\hat{\mathbf{v}} \triangleq \mathrm{vec}([\mathbf{V}]_{2:L+1,2:L+1})$, the second term of (33e) can be expressed as

$$\sigma_{\mathrm{t}}^2\mathrm{Tr}(\mathbf{\Theta}^H\mathbf{\Lambda}(\theta_{\mathrm{h}},\theta_{\mathrm{v}})\mathbf{\Theta}\tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\mathbf{W}^H\mathbf{A}^H\tilde{\mathbf{G}}_t^H\mathbf{\Theta}^H\mathbf{\Lambda}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}})\mathbf{\Theta}) \triangleq \|\mathbf{N}_2\hat{\mathbf{v}}\|_2^2, \tag{72}$$

where $\mathbf{N}_2 = \sigma_{\mathrm{t}}\mathbf{L}_1\,\mathrm{Diag}(\mathrm{vec}(\mathbf{\Lambda}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}})))$, and positive semidefinite matrix $\mathbf{I}_L \otimes (\tilde{\mathbf{G}}_t\mathbf{A}\mathbf{W}\mathbf{W}^H\mathbf{A}^H\tilde{\mathbf{G}}_t^H) = \mathbf{L}_1\mathbf{L}_1^{\mathrm{H}}$. The third term of (33e) can be expressed as

$$\sigma_{\mathrm{t}}^2\sigma_{\mathrm{z}}^2\mathrm{Tr}(\mathbf{\Theta}^H\mathbf{\Lambda}(\theta_{\mathrm{h}},\theta_{\mathrm{v}})\mathbf{\Theta}\mathbf{\Theta}^H\mathbf{\Lambda}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}})\mathbf{\Theta}) \triangleq \|\mathbf{N}_3\hat{\mathbf{v}}\|_2^2, \tag{73}$$

where $\mathbf{N}_3 = \sigma_{\mathrm{t}}\sigma_{\mathrm{z}}\,\mathrm{Diag}(\mathrm{vec}(\mathbf{\Lambda}^H(\theta_{\mathrm{h}},\theta_{\mathrm{v}})))$. The rest term of (33e) can be formulated as

$$\mathrm{Tr}(\mathbf{\Theta}^H\mathbf{\Theta}) = \mathrm{Tr}(\hat{\mathbf{V}}). \tag{74}$$

Therefore, the acitve RIS transmit power constraint can be expressed as

$$\mathrm{Tr}(\mathbf{N}_1\hat{\mathbf{V}}) + \|\mathbf{N}_2\hat{\mathbf{v}}\|_2^2 + \|\mathbf{N}_3\hat{\mathbf{v}}\|_2^2 + 2\sigma_{\mathrm{z}}^2\mathrm{Tr}(\hat{\mathbf{V}}) \leq P_{\mathrm{RIS}}. \tag{75}$$

Due to the difficulty in solving the original optimization problem with equality constraints, we adopt the PDD framework for processing. The augmented Lagrangian function of optimization problem is formulated as

$$\max_{\mathbf{V},\mathbf{V}_1,\xi}\ \xi + \frac{1}{2\rho}\|\mathbf{V}-\mathbf{V}_1\|_F^2 + \Re\left\{\mathrm{Tr}(\mathbf{\Lambda}_g^H(\mathbf{V}-\mathbf{V}_1))\right\} \tag{76a}$$

$$\text{s.t. } \tilde{R}_{\mathrm{sec},k} \geq \xi, \forall k \in \mathcal{K}, \tag{76b}$$

$$\mathrm{rank}(\mathbf{V}) = 1, \tag{76c}$$

$$[\mathbf{V}]_{1,1} = 1, [\mathbf{V}]_{l,l} \leq \beta_{\max}^2, l = 2,\ldots,L+1, \tag{76d}$$

$$\mathbf{V} \succeq 0, \tag{76e}$$

$$(67), (69), (75), \tag{76f}$$

where $\mathbf{\Lambda}_g$ is the Lagrangian dual variable. The PDD algorithm consists of two iteration loops, with its inner alternatively updating $\mathbf{V}$ and $\mathbf{V}_1$, and its outer layer selectively adjusting ρ or $\{\mathbf{\Lambda}_g\}$.

a) Outer Layer Update: The inner layer alternatively updates $\mathbf{V}$ and $\mathbf{V}_1$ until convergence is reached. For the outer layer iteration, the penalty parameter ρ^{-1} or the dual variable $\{\mathbf{\Lambda}_g\}$ is selectively updated depending on the residual value $\|\mathbf{V}-\mathbf{V}_1\|_\infty$, as detailed in the sequel.

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If $\|\mathbf{V}-\mathbf{V}_1\|_\infty$ is smaller than a threshold $\eta^{(s)}$, where the threshold sequence $\{\eta^{(s)}\}$ is diminishing, i.e., $\eta^{(s)} \to 0$ as $s \to \infty$, then we update the dual Lagrangian variable Λ_g as:

$$\mathbf{\Lambda}_g^{(s+1)} = \mathbf{\Lambda}_g^{(s)} + \rho^{-1}(\mathbf{V}^{(s)} - \mathbf{V}_1^{(s)}), \tag{77}$$

otherwise, it indicates that the constraint $\mathbf{V} = \mathbf{V}_1$ is currently far from being reached. In this case, we adjust the penalty coefficient ρ as follows

$$\rho^{(s+1)} = c \cdot \rho^{(s)}, \tag{78}$$

where c is a constant. The above update indeed inflates the penalty coefficient ρ^{-1}, thus forcing the constraint $\mathbf{V} = \mathbf{V}_1$ to be better satisfied in the subsequent iterations.

b) Inner Layer Update: The corresponding optimization problem is represented as

$$\max_{\mathbf{V},\mathbf{V}_1,\xi} \ \xi + \frac{1}{2\rho}\|\mathbf{V}-\mathbf{V}_1\|_F^2 + \Re\left\{\mathrm{Tr}(\mathbf{\Lambda}_g^H(\mathbf{V}-\mathbf{V}_1))\right\} \tag{79a}$$

$$\text{s.t. } (76b)-(76e), (67), (69), (75). \tag{79b}$$

Similar to the Problem (48), there exists the rank-one constraint (76c) that difficult to directly optimize in a convex method. When we ignore the constraint (76c), the original optimization Problem (79) is a convex second order cone programming Problem and using CVX solvers to solve.

C. Proposed D3QN-TD3 Method

For the MA position optimization, it is a mixed integer nonlinear programming problem, so the non-convex Problem (33) is difficult to solve using conventional methods. Therefore, DRL is introduced to solve the problem. DRL, as a powerful nonlinear optimization tool, can explore complex and high-dimensional state spaces and gradually optimize the solution to the problem through learning strategies. This section proposes a hybrid DRL framework to address the joint optimization problem of BS beamforming matrix, RIS phase shift, and MA position selection. Due to the mixed discrete-continuous action space, conventional DRL methods struggle to solve this problem. The D3QN-TD3 algorithm decomposes the entire ISAC secure transmission problem into an external Markov Decision Process with discrete action space solved by D3QN, and an internal Markov Decision Process with continuous action space solved by TD3. Specifically, D3QN handles the discrete actions for antenna position selection, while TD3 manages the continuous actions for beamforming and RIS phase shifts. The hybrid D3QN-TD3 model undergoes centralized training through a unified reward signal to coordinate both discrete and continuous policies. After training completion, the agent can execute policies in a decentralized manner to minimize communication overhead. Therefore, a D3QN-TD3 algorithm is introduced to enhance training stability and accelerate training speed.

1) Reinforcement Learning Process: To learn from the unknown environment, the agent needs to balance exploration and exploitation. That is to say, the agent will explore unknown knowledge by taking action randomly and exploit the learned knowledge by greedily taking action with the maximum Q-value. Herein, the objective of agent is to maximize the long-term cumulative reward for the MA position selection policy π.

At time slot n, given state $\boldsymbol{s}_n$ and action $\boldsymbol{a}_n$, the Q-function with policy π is defined as

$$Q(\boldsymbol{s}_n, \boldsymbol{a}_n) = \mathbb{E}_\pi\left[R_n | \boldsymbol{s}_n, \boldsymbol{a}_n\right]. \tag{80}$$

The optimal policy π^* can be obtained by maximizing $Q(\boldsymbol{s}_n, \boldsymbol{a}_n)$ as follows:

$$\pi^* \leftarrow \underset{\boldsymbol{a}_n \sim \pi}{\mathrm{argmax}} Q(\boldsymbol{s}_n, \boldsymbol{a}_n). \tag{81}$$

According to the Bellman criterion, the optimal Q-value is estimated as

$$Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n) = \boldsymbol{r}_n + \gamma \max Q^*\left(\boldsymbol{s}_{n+1}, \boldsymbol{a}_{n+1}\right), \tag{82}$$

which can be updated by

$$Q(\boldsymbol{s}_n, \boldsymbol{a}_n) \leftarrow (1-\alpha)Q(\boldsymbol{s}_n, \boldsymbol{a}_n) + \alpha Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n), \tag{83}$$

where α is the learning rate.

Theoretically, we can derive π^* by iteratively updating (83). However, due to the dynamic and random nature of the environment, iteratively determining π^* within an exploding state space is impractical. Therefore, integrating deep learning with reinforcement learning to address exploding state spaces represents the most effective solution currently available.

2) D3QN and TD3 Algorithm a) D3QN: The D3QN algorithm integrates DDQN and dueling DQN to improve the training performance of MDP problems. Specifically, the dueling DQN architecture employs two sets of fully connected layer networks in place of the traditional single neural network, enabling parallel estimation of both the state value function and the advantage function. Furthermore, the dual network mechanism incorporates an evaluation network and a target network, which mitigates the coupling between the action selection and the evaluation of action values, thereby addressing the prevalent issue of overestimation found in conventional DQN algorithms.

Due to channel time-varying characteristics and high-dynamic antenna adjustment strategies, the state space exhibits exponential growth, causing the Q-values stored in the Q-table to surge dramatically. This significantly reduces the learning efficiency of MA discrete optimal position selection, transmit beamforming, and RIS deployment strategies. To address this, we employ a deep neural network (DNN) with interconnected weighted neurons to approximate the Q-values, i.e.,

$$Q(\boldsymbol{s}_n, \boldsymbol{a}_n) \approx Q(\boldsymbol{s}_n, \boldsymbol{a}_n)|\theta), \tag{84}$$

where θ denotes the model parameter of DNNs.

As $Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n)$ increases with $Q(\boldsymbol{s}_n, \boldsymbol{a}_n)$ increasing, there will be oscillations and divergence. To address this problem, we employ the double architecture with estimation Q-network and target Q-network to approximate $Q(\boldsymbol{s}_n, \boldsymbol{a}_n)$ and $Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n)$, respectively. The estimation Q-network and target Q-network employ the same structure but with different parameters to generate Q-values. Specifically, at time t, the estimation Q-network utilizes the state $\boldsymbol{s}_n$ to generate the Q-value $Q(\boldsymbol{s}_n, \boldsymbol{a}_n)$ and the action $\boldsymbol{a}_n$, where θ represents the parameter of the estimation

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Q-network. Similarly, at time $n+1$, the target Q-network utilizes the state $\boldsymbol{s}_{n+1}$ to generate the Q-value $Q^*(\boldsymbol{s}_{n+1}, \boldsymbol{a}_{n+1}|\theta')$ and the action $\boldsymbol{a}_{n+1}$, where θ' represents the parameter of the target Q-network. According to (82), we have

$$Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta') = r_p(t) + \gamma \max Q^*(\boldsymbol{s}_{n+1}, \boldsymbol{a}_{n+1}|\theta'). \quad (85)$$

In this way, the temporal difference error at time n is calculated as

$$\delta = Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta') - Q(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta). \quad (86)$$

With (86), we employ stochastic gradient descent (SGD) method to update θ as

$$\nabla_\theta L(\theta) = \mathbb{E}\left[\delta \nabla_\theta Q(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta)\right], \quad (87)$$

where

$$L(\theta) = \mathbb{E}\left[(Q^*(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta') - Q(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta))^2\right], \quad (88)$$

is the loss function of the estimation Q-network. By updating θ with θ' for each round training, we can avoid oscillation and divergence.

On this basis, we use action $\boldsymbol{a}_n$ with the maximum Q-value in the estimation Q-network to update action $\boldsymbol{a}_{n+1}$ with the maximum Q-value in the target Q-network, i.e.,

$$Q^*(\boldsymbol{s}_{n+1}, \boldsymbol{a}_{n+1}|\theta') = Q^*(\boldsymbol{s}_{n+1}, \text{argmax} Q(\boldsymbol{s}_{n+1}, \boldsymbol{a}_{n+1}|\theta)|\theta'), \quad (89)$$

which helps mitigate the overestimation of Q-values that can arise from greedy exploitation in an expansive state space.

Additionally, we utilize a dueling network to evaluate the state, as certain computational decisions may be equivalent, rendering the state either favorable or unfavorable regardless of the action chosen within the expansive state space. Specifically, we use the average Q-value $\bar{Q}(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta)$ to estimate states, which as follows

$$\bar{Q}(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta) = V(\boldsymbol{s}_n|\theta) + \left(A(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta) - \frac{1}{|\mathcal{A}|}\sum_{\boldsymbol{a}_{n+1}\in\mathcal{A}} A(\boldsymbol{s}_n, \boldsymbol{a}_{n+1}|\theta)\right), \quad (90)$$

where $\mathcal{A}$ is the dimension of the action space. $V(\boldsymbol{s}_n|\theta)$ and $A(\boldsymbol{s}_n, \boldsymbol{a}_n|\theta)$ represent the state value function and action advantage function, respectively.

b) TD3: The TD3 algorithm employs two identical networks to estimate Q values, utilizing the smaller Q value as the update target to address the issue of overestimation. Specifically, the architecture of the TD3 network comprises two components: the actor network and the critic network. ϑ and ϑ^- denote the network parameters of the current actor network and the target actor network, respectively. τ_1 and τ_1^- denote the network parameters of the first current critic network and the first target critic network, respectively. τ_2 and τ_2^- denote the network parameters of the second current critic network and the second target critic network, respectively.

In addition, the TD3 algorithm solves the problem of overestimation bias through the following three techniques to improve performance. The first technique adds noise to the target action and moderately smooths the target strategy to alleviate overfitting issues. The second technique is to reduce double Q learning. In the TD3 network, both the critic target network and the current network contain two twin networks, which can simultaneously learn two Q-value functions and select the smaller Q-value as the target Q-value for the Bellman error function. The third technique is delayed policy update, which sets the critic network to update every two times and the actor network to update again, making the critic network more stable and reducing errors before updating the actor network.

Regarding the update method for the actor network, specifically, the TD3 network uses the Q-values generated by the first critic network to update the actor network.

For the critic network, a smaller Q value is obtained by comparing two current critic networks, combined with the reward to generate the target value y, which is expressed as

$$y = \boldsymbol{r}_{n+1} + \gamma \min_{i=1,2} Q_i\left(\boldsymbol{s}_{n+1}, \boldsymbol{a}; \tau_i^-\right)|_{\boldsymbol{a}=\pi(\boldsymbol{s}_{n+1}, \vartheta^-)}. \quad (91)$$

For the experience replay buffer, randomly select X tuples $\{\boldsymbol{s}_m, \boldsymbol{a}_m, \boldsymbol{r}_m, \boldsymbol{s}_{m+1}\}$ to train the Critic network. The parameter weight τ_i of the Critic network is updated by minimizing the loss function $L(\tau_i)$, where the loss function is expressed as

$$L(\tau_i) = \frac{1}{X}\sum_m \left[y - Q_i(\boldsymbol{s}_m, \boldsymbol{a}_m; \tau_i)\right]^2. \quad (92)$$

The algorithm workflow framework based on D3QN-TD3 is in detail summarized in Algorithm 1. At the beginning of each training iteration, the agent observes and collects environmental channel information to obtain the initial system state $\boldsymbol{s}_0$. At time step n, the state is input into the D3QN network to obtain the discrete transmit/receiving MA selection $\mathbf{A}_n$ and $\mathbf{B}_n$. The TD3 network is used to obtain the continuous secondary BS beamforming $\mathbf{W}_n$ and RIS active phase shift $\boldsymbol{\Theta}_n$. By executing $\{\mathbf{A}_n, \mathbf{B}_n, \mathbf{W}_n, \boldsymbol{\Theta}_n\}$ and interact with the environment to obtain reward $\boldsymbol{r}_n$ and the next state $\boldsymbol{s}_{n+1}$, and continuously train until the algorithm converges.

3) Computational Complexity Analysis: To evaluate the efficiency of the proposed solution, we mainly analyze the computational complexity of the Algorithm 1. Firstly, we analyze the computational complexity of the optimization of transmit beamforming and RIS phase shift. It is obvious that the update for $\mathbf{W}$ and $\boldsymbol{\Theta}$ dominate the computational cost, whose complexities are of order $\mathcal{O}\{M^{4.5}\}$ and $\mathcal{O}\{C_1C_2(M^4\sqrt{L}+M^3)\}$, respectively, where C_1 and C_2 is the number of the loops of outer-layer and inner-layer in PDD framework, respectively [85], [86]. Thus, the overall complexity of the AO-based algorithm is of order $\mathcal{O}\{M^{4.5}+C_1C_2(M^4\sqrt{L}+M^3)\}$. Next, we analyze the computational complexity of D3QN-TD3 algorithm, which adopts offline training and online execution policy. In both offline training and online execution, the mapping from states to actions is realized by DNNs, whose computational complexity is given as $\mathcal{O}_J = \mathcal{O}(\sum_{j=1}^{J} o_j o_{j+1})$, where J and o_j are the number of layers and the number of neurons at the j-th layer, respectively. In offline training, with T training times, N training steps, and Z training data, the computational complexity is calculated as $\mathcal{O}(\sum_{j=1}^{J} o_j o_{j+1} TNZ)$. As the training process is performed

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Algorithm 1: AO-based D3QN-TD3 Algorithm for Solving Problem.

Initialization: Experience memory B, θ and θ' of the evaluation the Q-networks $Q(\cdot)$ and target network $Q^*(\cdot)$, the training actor network parameter ϑ and the target actor network parameter ϑ^-, the first training critic network parameter τ_1 and the first target critic1 network parameter τ_1^-, the second training critic network parameter τ_2 and the second target critic network parameter τ_2^-.
for each episode **do**
  Initialize the system state $\boldsymbol{s}_0$;
  repeat
  Obtain $\mathbf{J}$ by solving (44);
  Obtain $\mathbf{W}$ by solving (54);
  Obtain $\boldsymbol{\Theta}$ by solving (76);
  until Problem (43) in coverged
  for $n = 0, 1, \ldots, N$ **do**
    Collect current $\mathbf{W}$, $\boldsymbol{\Theta}$, $\mathbf{A}$, and $\mathbf{B}$ to observe state $\boldsymbol{s}_n$.
    Obtain action $\boldsymbol{a}_n = \{\boldsymbol{a}_n^1, \boldsymbol{a}_n^2\}$ from D3QN and actor network in TD3, respectively;
    Store $\{\boldsymbol{s}_n, \boldsymbol{a}_n, \boldsymbol{r}_n, \boldsymbol{s}_{n+1}\}$ in experience memory B;
    $\boldsymbol{s}_{n+1} = \boldsymbol{s}_n$;
    Select a random mini-batch of samples from B;
    Perform SGD to update D3QN network parameter θ;
    Obtain the loss function for the training critic network according to (91);
    Update the actor policy by the sampled policy gradient;
    Update training actor network parameter ϑ and training critic network parameters τ_1 and τ_2, respectively;
    Update D3QN target network parameter θ';
    Update target critic network parameter ϑ^- and target actor network parameters τ_1^- and τ_2^-, respectively;
  end for
end for

offline, we can realize real-time online execution for the complex industrial tasks, wherein the computational complexity is only $\mathcal{O}_J$.

V. Simulation Results

In this section, the simulation results validate the performance of MA-enabled secure transmission for active RIS-aided ISAC systems. Communication users and the eavesdropping target are equipped with a single FPA. The transmit and receive region for the MA is modeled as a 2D square, and the minimum distance interval between each pair of MA is defined as $D_{1,\min} = D_{2,\min} = \lambda/2$, the transmission area size is set to A, where $\lambda = 0.105$ m represents the carrier wavelength. The locations of ISAC BS and active RIS are set to (0 m, 0 m, 3 m) and (0 m, 20 m, 3 m), respectively. The communication users are located in a circle with the center at (10 m, 20 m, 0 m) and a radius of 5 m. A geometric channel model is adopted, where the numbers of transmit and receive paths are equal, i.e. $L_t = L_r = L_t^k = L_r^k = 6$, $\forall k$ [87]. According to reference [65], the path response matrix $\boldsymbol{\Sigma}_k$ and $\boldsymbol{\Sigma}$ are a diagonal matrix, with each diagonal element following a distribution $\mathcal{CN}(0, c_0 d^{-\alpha}/L_0)$, where $c_0 = -40$ dB denotes the channel gain at 1m and $\alpha = 2.8$ is the pathloss exponent, respectively. The elevation angle and azimuth angle (AoA/AoD) of the user/target (RIS) are treated as independent and identically distributed random variables, each following a uniform distribution. To simplify the analysis, it is assumed that the noise variance of all communication and radar devices, as well as RIS, is equal, i.e., $\sigma_k^2 = \sigma_{\mathrm{r}}^2 = \sigma_{\mathrm{z}}^2 = -90$ dBm, where $1 \le k \le K$. Unless otherwise specified, the number of users, the number of transmit MA and receive MAs, the transmit power budget of BS, the communication SINR requirement, the number of active RIS elements, the power threshold of active RIS, and the CRB threshold are set to $K = 4$, $M = N = 6$, $P_{\mathrm{BS}} = 30$ dBm, $\Gamma = 5$ dB, $L = 128$, $P_{\mathrm{RIS}} = 10$ dBm, and $\delta = 0.01$, respectively.

The experiments took place on a Windows platform equipped with an Intel Core i7 processor, a GeForce 4070 graphics card, and 16 GB of RAM. In the proposed D3QN-TD3 algorithm framework, D3QN network contains three hidden layers, and actor network and critic network in TD3 have two hidden layers. The learning rate of D3QN is set as 0.01, the learning rates of the actor network and the critic network are set as 0.001 and 0.002, respectively. The discount factor is set as $\gamma = 0.99$, and the experience replay buffer is $L = 20000$.

Furthermore, we consider five other benchmark approaches for comparison with our proposed method, which are listed as follows:

- *D3QN-TD3:* Without considering AO algorithm for the optimization of transmission beamforming and RIS phase shifts, directly apply the method combining D3QN and TD3 for the MA-assisted RIS-aided system.
- *FPA:* The transmitting and receiving antennas are fixed in determined positions to achieve better SR performance.
- *Random position antenna (RPA):* The transmitting and receiving antennas are randomly distributed within the available area.
- *Passive RIS:* Passive RIS replaces active RIS to serve the MA-enabled system and ensure SR performance.
- *Random phase shift (RPS):* Each element on the active RIS has an amplitude of unity but a random phase shift following a uniform distribution within the range $[0, 2\pi)$.

Fig. 3 illustrates the convergence behavior of the proposed AO-based D3QN-TD3 method compared to D3QN-TD3 benchmark scheme. As the number of training rounds increases, each curve gradually stabilizes, indicating that the algorithm converges over time. Notably, the reward value of the AO-based D3QN-TD3 method continues to increase and stabilizes after 90 iterations, whereas the D3QN-TD3 algorithm requires 150 iterations to achieve convergence. This demonstrates that the AO-based D3QN-TD3 algorithm exhibits significant advantages

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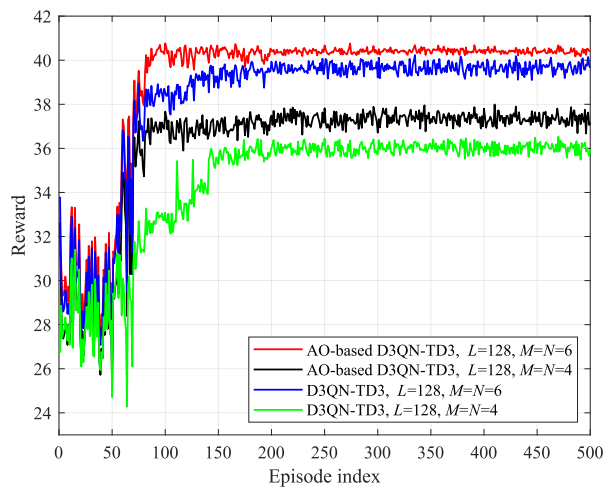

Fig. 3. Convergence behavior of proposed AO-based D3QN-TD3 method.

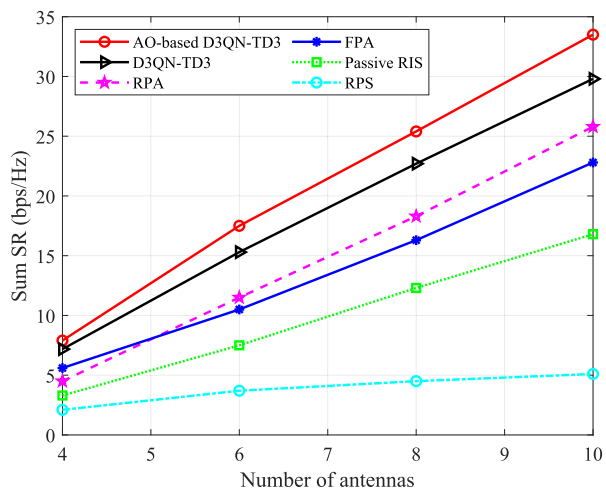

Fig. 4. Sum SR versus the number of antennas M.

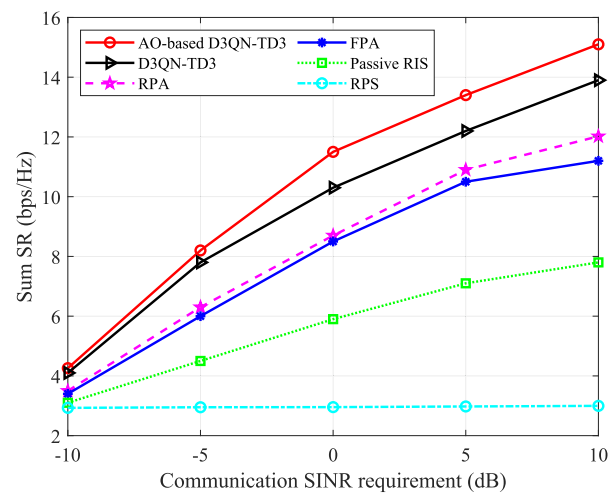

Fig. 5. Sum SR versus the communication SINR requirement Γ.

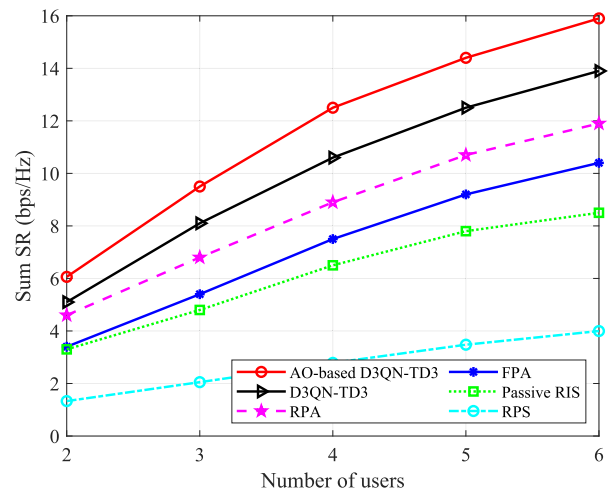

Fig. 6. Sum SR versus the number of users K.

in both stability and training speed compared to the D3QN-TD3 algorithm. The accelerated initial training process of the proposed algorithm contributes to this performance, as evidenced by the increasing reward value with the addition of multiple MAs. This correlation arises because the reward value is associated with the SR, thus, as the system utilizes more MAs, the reward value correspondingly increases.

Fig. 4 illustrates the sum SR behavior of versus the number of antenna. As the number of training rounds increases, each curve gradually stabilizes, indicating that the algorithm converges over time. Notably, the reward value of the AO-based D3QN-TD3 method continues to increase and stabilizes after 90 iterations, whereas the D3QN-TD3 algorithm requires 150 iterations to achieve convergence. This demonstrates that the AO-based D3QN-TD3 algorithm exhibits significant advantages in both stability and training speed compared to the D3QN-TD3 algorithm. The accelerated initial training process of the proposed algorithm contributes to the performance, as evidenced by the increasing reward value with the addition of multiple MAs. The correlation arises because the reward value is associated with the SR, thus, as the system utilizes more MAs, the reward value correspondingly increases.

Fig. 5 depicts the sum SR behavior comparison for various methods against the communication SINR requirement. The results demonstrate that this scheme consistently outperforms all benchmark methods under various communication SINR requirement Γ. This indicates that when Γ is fixed, the AO-based D3QN-TD3 method achieves a higher lower bound on the secrecy rate, thereby significantly enhancing system security. For example, $\Gamma = 10$ dB, the AO-based D3QN-TD3 method achieves a secrecy rate for all users is 15.13bps/Hz. However, under constrained transmission power budgets and radar estimation constraints, the secure transmission power for communication information imposes an upper limit, resulting in a slight decline in the performance growth trend.

Fig. 6 demonstrates the sum SR behavior comparison for various methods against the number of users K. As the number of users increases, the Sum Secrecy Rate shows a significant upward trend. It is also demonstrated that the AO-based D3QN-TD3 algorithm exhibits optimal security performance in the

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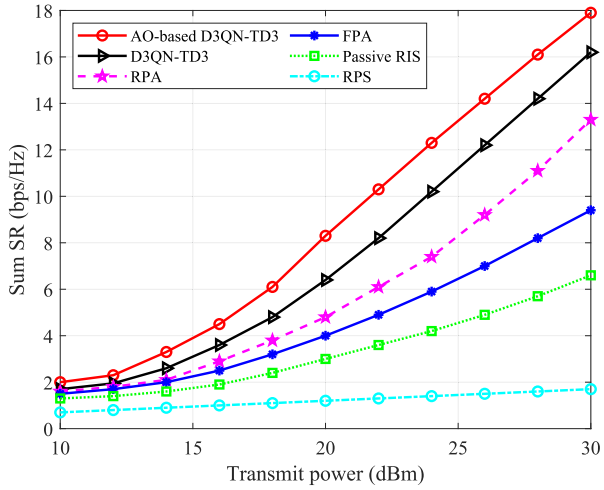

Fig. 7. Sum SR versus the transmit power P_{BS}.

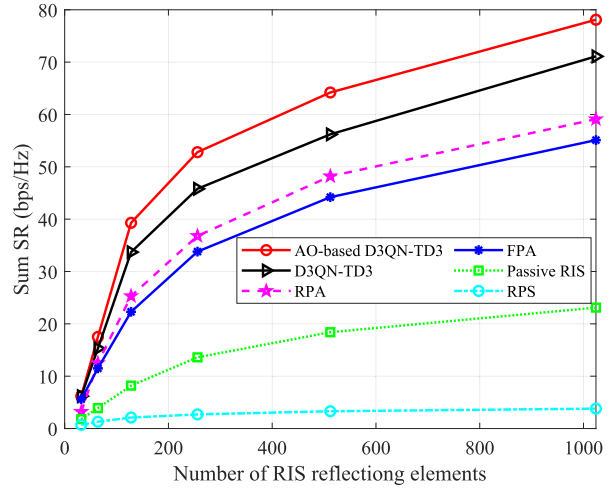

Fig. 8. Sum SR versus the number of RIS elements L.

MA-enabled Active RIS system. This is because the system can effectively suppress information leakage to eavesdroppers through beamforming and antenna movement, resulting in the cumulative increase of the total secrecy rate with the growing number of served users. Furthermore, this also reflects the high-precision beamforming capability of the AO algorithm, which minimizes information leakage and compensates for the accuracy limitations of pure DRL in handling continuous high-dimensional beam optimization.

Fig. 7 illustrates the sum SR behavior comparison for various methods against the transmit power P_{BS}. As P_{BS} increases, the sum SR performance exhibits a rapid growth trend. Furthermore, the ISAC system utilizing an active RIS solution demonstrates performance improvements of approximately 110% and 150% over "passive RIS" and "RPS" schemes, respectively. This highlights the critical importance of deploying RIS elements in ISAC systems to prevent malicious eavesdropping. Additionally, the MA scheme effectively leverages spatial DoF, resulting in approximately 30% performance gains over the FPA scheme across all scenarios. This enhancement can be not only attributed to the higher transmission power enhancing the anti-interference ability of the signal in complex environments, but also the capture of higher spatial DoF by MA array.

Fig. 8 illustrates the sum SR behavior comparison for various methods against the communication SINR requirement. The results show that when the number of reflectors increased from 64 to 128, the system's overall SR improved by 21.87 bps/Hz. This indicates that appropriately increasing the number of RIS elements can substantially enhance the SR. Furthermore, under identical conditions of the number of RIS elements, the introduction of active RIS scheme consistently outperforms the "Passive RIS". This is attributable to the significant capacity gains achieved by the active RIS system in mitigating the effects of multipath fading.

In Fig. 9, we analyze the relationship between sum SR behavior and the normalized moving region area of MAs. It can be observed that the total movement distance of MAs increases linearly with the size of the movement area, and the proposed scheme can significantly enhance security performance when compared to the "FPA" and "RPA" schemes.

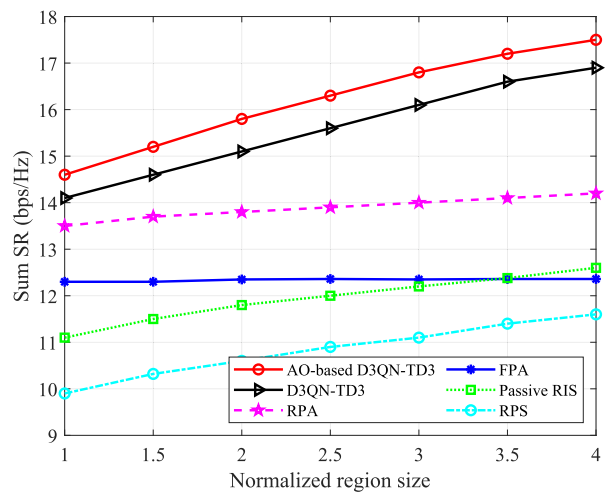

Fig. 9. Sum SR versus the normalized moving region size of antennas A/λ.

For example, when the normalized moving region area equals 4, the sum SR achieved by the "FPA" scheme is 14.21 bps/Hz while that achieved by the proposed scheme is approximately 17.56 bps/Hz. The phenomenon occurs because a larger reception area provides greater flexibility for optimizing MA positions, enabling them to relocate to areas with improved channel conditions and thereby enhancing system security. However, it is important to note that the performance improvement derived from expanding the movable area gradually diminishes as the normalized moving region area increases.

Finally, Fig. 10 describes the relationship between sum SR behavior and the normalized moving region area of MAs. Compared to the "PPA" solution, the MA approach capitalizes on its spatial multiplexing gain advantage to achieve superior performance in secure transmission, irrespective of radar functionality. As the CRB threshold δ increases, the transmission of safe rates tends to stabilize. The phenomenon occurs because progressively relaxing constraints on radar performance allows for a greater allocation of transmission power to radar estimation, consequently reducing the power available for base station information transmission.

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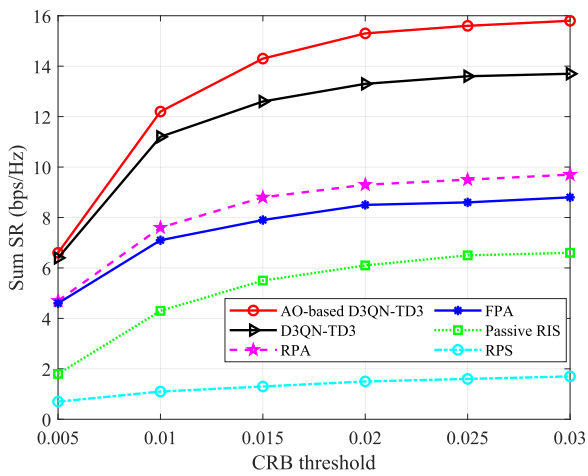

Fig. 10. Sum SR versus the CRB threshold δ.

VI. Conclusion

In this paper, we studied a novel MA-enabled active RIS-aided ISAC systems to enhance the secure transmission. The objective of this study was to maximize the minimum SR by the joint design of transmit beamforming and phase shift of RIS, as well as the MA discrete location. Given the non-convexity of the objective function and the difficulty in solving it using traditional methods, we developed an AO-based D3QN-TD3 method to handle the hybrid action space issue. Specifically, the effective integration of AO optimization with the D3QN-TD3 algorithm framework resolves issues of algorithmic processing difficulties and getting stuck in local optima. Numerical results indicated that the proposed MA-assisted ISAC system significantly outperforms conventional FPA scheme in terms of secure transmission. In addition, it is worth noting that the algorithm we proposed is highly effective in practical scenarios involving discrete antenna positioning. In the future, we will consider comparing the performance upper limits of discrete MA optimization and continuous MA optimization for ISAC systems. Additionally, we will focus on investigating the impact of 6D MA-assisted ISAC systems on secure transmission.

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